

Sharp Tail Asymptotics and Efficient Rare-Event Simulation for Independent and Dependent Regularly-Varying Factor Models: Latent Shocks, Spectral Measures, Hidden Cones, and Financial Evidence*

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Abstract

We study rare-event probabilities and efficient simulation for finite sums of regularly varying factor-loss contributions. The first layer of the paper gives sharp independent-factor theory. For independent, non-identically distributed summands satisfying pointwise tail equivalence, we prove the exact finite- portfolio catastrophe principle for the maximum and the corresponding single- big-jump equivalence for the sum under the stated truncation and left-tail conditions. Under direct second-order tail equivalence, local-shift regularity, and finite first moments, we derive a corrected expansion for $\mathbb{P}(S_N > x)$. The mean-shift correction is a leave-one-out term, $\mu_{-i} = \sum_{j \neq i} \mathbb{E}X_j$, because the large summand supplies the threshold crossing while the remaining portfolio supplies the first-order translation. This form passes the $N = 1$ consistency check and produces explicit Pareto/Lomax benchmarks.

The second layer develops a staged extension path for non-independent factors. For arbitrary dependence, we prove an exact disjoint-partition conditional Monte Carlo identity in which the independent marginal survival function is replaced by the conditional tail $\mathbb{P}(X_i > T_i \mid X_{-i})$. We then specialize this identity to four dependence structures that cover the main financial use cases: latent independent shocks and common shocks, independent blocks with dependent within-block components, copula and vine-copula conditional tails, and multivariate regular variation with a spectral/radial representation. The resulting hierarchy replaces the independent “one large coordinate” principle by “one large latent shock”, “one large block”, and ultimately “one large spectral vector shock”. Hidden regular variation is formulated as a second- order joint-tail correction when ordinary multivariate regular variation puts first-order mass on the coordinate axes but slower joint-tail cones remain empirically material.

Building on these asymptotics, we analyze summed CdMC, single-index stratified CdMC, Pareto/Lomax closed forms, and centered control-variate estimators. In the independent case

*The methods of this paper are implemented in the open-source **FactorTail** library, available at <https://github.com/osolari/FactorTail>.

the summed CdMC estimator is unbiased and has the scale- invariant bounded-relative-error bound $N^\alpha - 1$. In dependent settings the same partition gives unbiasedness, while bounded-relative-error guarantees require model-specific conditional-tail, block, latent-shock, or radial-envelope assumptions. The manuscript includes full proof templates for the dependent identities and for the latent-shock, block, copula, and spectral/radial extensions.

The numerical and empirical sections are written as a reproducible long-form analysis plan. The simulation design covers independent, common-shock, block- dependent, Gaussian-copula, Student- t -copula, vine-copula, multivariate- regular-variation, and hidden-regular-variation data-generating processes. The real-data design covers CRSP portfolio returns and Fama–French factors, rolling factor fits, one-sided tail estimation, tail-dependence diagnostics, empirical spectral measures, hidden-cone diagnostics, VaR/ES forecasting, conditional-Monte-Carlo estimator comparisons, and formal backtesting. All figures and tables are placeholders tied to explicit result schemas; planned numeric values are displayed as `xx` until generated by the frozen empirical pipeline.

The paper positions itself as a sharpening and extension of the independent-factor regularly-varying rare-event framework of Pourbabaee and Solari [2019], integrating tools from heavy-tailed rare-event simulation [Asmussen and Kroese, 2006, Hartinger and Kortschak, 2009], multivariate extremes [Hult et al., 2005, Resnick, 2007], hidden regular variation [Ledford and Tawn, 1996, Maulik and Resnick, 2005], and dependent rare-event simulation [Blanchet and Rojas-Nandayapa, 2011, Cheng et al., 2025] into a single staged estimator-and-diagnostic pipeline. An annotated literature synthesis appears in Section 2. The methods are implemented in the open-source `FactorTail` library (<https://github.com/osolari/FactorTail>), which provides reference implementations of every estimator, diagnostic, and reproducibility artifact discussed in this paper.

Keywords: rare-event simulation; regularly varying tails; multivariate regular variation; hidden regular variation; dependent factor models; conditional Monte Carlo; bounded relative error; vanishing relative error; Fama–French factor model; value at risk; expected shortfall.

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1 Introduction

1.1 Motivation

Heavy-tailed risk models are built around an asymmetry between ordinary fluctuations and rare losses. When the loss distribution is regularly varying, large portfolio exceedances are not well represented by Gaussian perturbations: they are typically generated by one extreme component, one extreme latent shock, one extreme block, or one extreme vector direction. The simplest version of this idea is the single-big-jump principle for independent heavy-tailed summands. Financial factor portfolios, however, rarely contain exactly independent factors. Market, size, value, profitability, investment, country, sector, credit, volatility, liquidity, and residual terms can co-move in their extremes. This paper therefore develops the independent theory carefully and then turns it into a staged framework for non-independent factor-loss models.

Let

$$S_N = \sum_{i=1}^N X_i$$

be the tail-risk target, where the X_i are signed one-sided loss contributions obtained from factor returns, portfolio exposures, and residuals. The independent baseline assumes

$$\mathbb{P}(X_i > x) \sim c_i \overline{G}(x), \quad \overline{G} \in \text{RV}_{-\alpha},$$

for a common reference tail and a finite number of summands. The leading question is the rare-event probability

$$\mu(x) = \mathbb{P}(S_N > x), \quad x \rightarrow \infty.$$

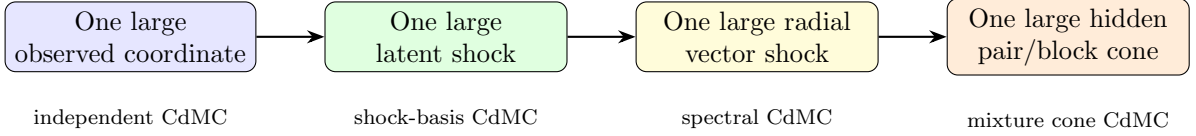


Figure 1: Hierarchy of big-jump mechanisms developed in the paper. The independent finite- N theory corresponds to axis mass, latent-shock models move the conditioning to independent underlying shocks, block models allow within-block joint extremes, multivariate regular variation integrates over a spectral measure, and hidden regular variation captures slower joint-tail cones that may be material in finite samples.

The computational question is how to estimate $\mu(x)$ accurately when crude Monte Carlo sees few or no exceedances.

The paper has two complementary objectives. First, it sharpens the independent finite- N theory and the conditional Monte Carlo estimators used by Pourbabaee and Solari [2019]. Second, it develops a concrete route from that independent theory to non-independent factor models. The dependent route is not presented as a single over-general theorem. Instead, it is organized by the source of tail dependence: arbitrary conditional tails, latent independent shocks, common shocks, independent blocks, copulas, multivariate regular variation, and hidden regular variation.

1.2 Core message

The core message is a hierarchy of rare-event mechanisms:

one large coordinate $\subset$ one large latent shock $\subset$ one large block $\subset$ one large spectral vector shock $\subset$ one large

The independent theory is the coordinate-axis special case. A common-shock factor model replaces several observed extreme coordinates by a single latent source. A block model permits multiple components to move together inside a sector, region, or factor cluster. A multivariate regularly varying model replaces the active coordinate set by a spectral measure over extreme directions. Hidden regular variation adds slower joint-tail cones that are invisible to the first-order limit but can dominate ordinary second-order marginal corrections.

1.3 Roadmap of assumptions

The theoretical results in this paper layer assumptions cumulatively rather than invoking a single global hypothesis. The independent baseline of Section 4 requires only common-reference regular variation, a non-empty active set, a deterministic tie-breaking rule, and measurability Assumptions 3.1 to 3.4. The second-order expansion adds direct second-order tail equivalence Assumption 3.5 and a finite first moment ($\alpha > 1$). The dependent conditional Monte Carlo identity of Section 5 relaxes independence to arbitrary dependence and replaces the marginal survival by the conditional kernel; bounded-relative-error guarantees in that section add model-specific conditions stated where they bind. The MRV extension of Section 6 adds multivariate regular variation of the loss-vector plus boundary continuity of the linear loss set under the limit measure. The hidden-regular-variation extension of Section 7 adds a hidden-cone second-order assumption. Appendix A consolidates the symbol inventory.

1.4 Contributions

Independent finite- N asymptotics. Section 4 proves the exact finite-portfolio catastrophe principle for the maximum and the corresponding single-big-jump equivalence for the sum. For the active set $\mathcal{A} = \{i : c_i > 0\}$, the leading term is

$$\mathbb{P}(S_N > x) \sim \sum_{i \in \mathcal{A}} \mathbb{P}(X_i > x) \sim \left(\sum_{i \in \mathcal{A}} c_i \right) \overline{G}(x).$$

The section also records finite-sample bounds that make the asymptotic constants usable as implementation checks.

Corrected second-order expansion. Under direct second-order tail equivalence, local-shift regularity, and finite first moments, Section 4 derives

$$\mathbb{P}(S_N > x) = C \overline{G}(x) + \overline{G}(x) A(x) \sum_{i \in \mathcal{A}} c_i d_i + \frac{\alpha \overline{G}(x)}{x} \sum_{i \in \mathcal{A}} c_i \mu_{-i} + o\{\overline{G}(x)(a(x) \vee x^{-1})\},$$

where $C = \sum_{i \in \mathcal{A}} c_i$, $a(x) = |A(x)|$, and $\mu_{-i} = \sum_{j \neq i} \mathbb{E}X_j$. The leave-one-out mean is essential: once coordinate i is the large component, the remaining portfolio supplies the local shift.

Dependence extension path. Section 5 proves that the same maximum-index partition gives an exact unbiased estimator under arbitrary dependence after replacing the marginal survival $\overline{F}_i$ by the conditional survival $\mathbb{P}(X_i > t \mid X_{-i})$. It then gives implementable specializations for latent independent shocks, common shocks, independent blocks, copulas, and vine copulas. Section 6 develops the multivariate-regular-variation extension in which the one-large-coordinate constant becomes a spectral-measure integral. Section 7 formulates hidden regular variation as a second-order joint-tail correction when ordinary MRV puts first-order mass on the coordinate axes but slower cones remain material.

Estimator family and efficiency criteria. Section 8 compares crude Monte Carlo, independent CdMC, dependent conditional-tail CdMC, latent-shock CdMC, block CdMC, spectral/radial CdMC, and control-variate variants. In the independent case, the summed CdMC estimator

$$Z^{\text{CdMC}}(x) = \sum_{i=1}^N \overline{F}_i((x - S_{-i}) \vee M_{-i})$$

is unbiased and has the scale-invariant bounded-relative-error bound $N^\alpha - 1$. In dependent settings, the same partition gives unbiasedness, but bounded-relative-error guarantees require conditional-tail, block, latent-shock, or radial-envelope assumptions.

Open-source reference implementation. All estimators, diagnostics, and reproducibility utilities developed in this paper are implemented in the **FactorTail** Python library, available at <https://github.com/osolari/FactorTail>. The library mirrors the manuscript's module structure: independent CdMC, dependent kernel CdMC, latent-shock CdMC, block CdMC, spectral/radial CdMC, hidden-cone diagnostics, and the full Fama–French / CRSP rolling

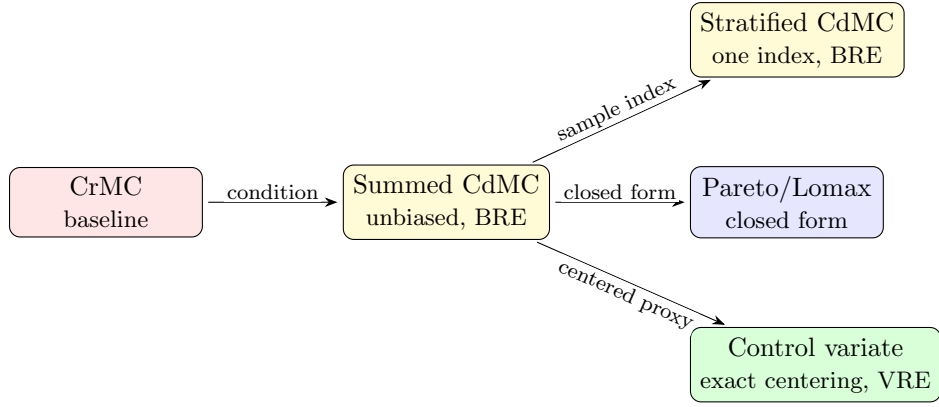


Figure 2: Conditional Monte Carlo estimator family. Summed CdMC analytically conditions on each candidate dominant summand. Stratified CdMC samples one conditioning index per replicate. Pareto/Lomax closed form is a specialization of CdMC, and the centered control-variate path uses exact or sample-split centering.

VaR/ES pipeline. Section references in the manuscript correspond to module paths in the library, and every algorithm in Appendix E has a one-to-one Python entry point.

Long-form empirical design (planned). Section 9 and Section 10 are written as complete protocols rather than short placeholders. The simulation design specifies data-generating processes, estimators-to-test, replication counts, success criteria, and reproducibility metadata for independent and dependent regimes. The real-data design specifies Fama–French factor data, CRSP returns, rolling factor estimation, one-sided tail fitting, dependence diagnostics, empirical spectral estimation, hidden-cone diagnostics, VaR/ES forecasting, estimator runtime and variance comparisons, and backtesting. Figures and tables in Section 9 and Section 10 are implementation placeholders with planned values marked **xx**; numerical performance and backtest outcomes are not asserted until generated outputs replace those entries.

1.5 Scope and status of empirical claims

The theorem sections establish mathematical claims under explicit assumptions. The simulation and real-data sections specify experiments, diagnostics, schemas, and replacement rules. Numeric performance, backtest success, and production runtime improvements are not asserted until generated outputs replace the **xx** entries. This separation is deliberate: the paper can be read as a theory paper, an estimator paper, and an empirical protocol without confusing planned results with observed results. The **FactorTail** library (<https://github.com/osolari/FactorTail>) supplies the implementation against which any future numeric results must be reproducible.

1.6 Organization

Section 2 places the paper in the literature. Section 3 introduces notation, assumptions, and the financial loss mapping. Section 4 gives the leading and second-order independent asymptotics. Section 5 develops exact dependent conditional Monte Carlo identities and practical dependence modules. Section 6 and Section 7 develop spectral and hidden-cone extensions. Section 8 compares estimators and efficiency guarantees. Section 9 specifies the planned synthetic experiments.

Section 10 gives the extensive real-data analysis plan. Section 11 concludes. Appendices collect notation, proofs, pseudo-code, reproducibility rules, dataset specifications, and experiment manifests.

2 Literature Review and Positioning

2.1 Heavy-tailed sums and the single-big-jump principle

The theoretical base is the large-deviation theory of subexponential and regularly varying sums. In the finite-dimensional setting, the key heuristic is that a high exceedance of a heavy-tailed sum is caused by one large term rather than by a Gaussian accumulation of many small terms. This idea appears in the standard treatments of regular variation and subexponentiality [Bingham et al., 1987, Embrechts et al., 1997, Resnick, 2007, Foss et al., 2013] and in random-walk large-deviation results such as Denisov et al. [2008]. The independent baseline of this paper is a finite, non-identically distributed version of that principle, with the additional goal of producing simulation estimators and financial risk forecasts.

2.2 Rare-event simulation for heavy tails

Conditional Monte Carlo for heavy-tailed sums was developed to avoid the poor performance of crude Monte Carlo at very small probabilities. Asmussen and Kroese [2006] introduced an influential estimator based on conditioning on all but a candidate large summand. Subsequent work studied extensions, efficiency refinements, and related estimators, including Chan and Kroese [2011], Hartinger and Kortschak [2009], Asmussen and Kortschak [2015], and Ghamami and Ross [2012]. General simulation criteria and rare-event efficiency concepts are reviewed in Asmussen and Glynn [2007], Glasserman [2004], and Juneja and Shahabuddin [2006]. The independent part of the present paper fits this line but emphasizes non-identical tail constants, signed financial contributions, exact active-set weights, finite-threshold concentration, and a corrected finite-mean second-order expansion.

2.3 Dependent heavy-tailed factor models

The closest conceptual predecessor for dependent factor models is Djehiche and Svensson [2007], who study large deviations for sums generated by a heavy-tailed factor structure and identify cases where factor and idiosyncratic terms contribute to the tail probability. The earlier factor-model rare-event work of Pourbabaee and Solari [2019] is the direct predecessor of the independent baseline. The present draft bridges those two lines by making the conditioning object explicit. Observed factor dependence can be handled by moving from observed coordinates to latent shocks, independent blocks, conditional copula kernels, or spectral directions.

Positioning relative to Pourbabaee–Solari (2019). Pourbabaee and Solari [2019] establish the first-order single-big-jump equivalence and a CdMC estimator for an independent, non-identically distributed finite-factor model. The present paper sharpens that result in four directions. First, the active-set weights c_i and the active-set constant C enter the limit theorem

and the BRE bound symmetrically; the BRE bound $N^\alpha - 1$ is proved here in scale-invariant form (Section 4). Second, the second-order expansion is corrected so that the local shift is supplied by the leave-one-out mean $\mu_{-i} = \sum_{j \neq i} \mathbb{E}X_j$ rather than the full portfolio mean; the resulting expansion passes an $N = 1$ specialization check that the prior expansion did not. Third, the dependent identity of Section 5 extends the same partition argument to arbitrary joint laws, recovering the independent case by setting the conditional kernel equal to the marginal survival. Fourth, the multivariate-regular-variation and hidden-regular-variation extensions of Sections 6 and 7 are new relative to the independent setting.

2.4 Dependent rare-event simulation

For dependent heavy-tailed random variables, the simulation literature includes conditional Monte Carlo for log-elliptical structures and importance-sampling methods under broader dependence. Blanchet and Rojas-Nandayapa [2011] propose asymptotically optimal algorithms for tail probabilities of sums of dependent random variables, including a conditional Monte Carlo algorithm for elliptical log variables and an importance sampler for general dependence under conditional simulation assumptions. Recent copula-based work such as Cheng et al. [2025] develops efficient importance sampling for copula models. The present paper differs by keeping the factor-portfolio structure, the regularly varying signed-tail constants, and the real-data VaR/ES workflow at the center. Its dependent kernel estimator is exact whenever the conditional survival is exact, while the shock, block, and spectral estimators choose more stable conditioning objects when coordinate kernels are inefficient.

2.5 Multivariate regular variation and hidden regular variation

Multivariate regular variation is the natural language for dependent extremes. Hult et al. [2005] extend heavy-tailed large deviations to multivariate regularly varying random walks and make precise the idea that a single large vector step drives extremal behavior. The spectral measure encodes whether extremes concentrate on coordinate axes, rays, faces, or larger angular regions. Samorodnitsky and Sun [2016] develop multivariate subexponential distributions and applications to ruin problems beyond the simplest independent marginal setting.

Hidden regular variation is relevant when ordinary MRV is asymptotically independent but joint-tail regions remain practically important. Maulik and Resnick [2004] characterize hidden regular variation and relate it to residual tail dependence ideas. Das et al. [2013] emphasize that classical MRV can understate probabilities of joint tail regions and that hidden regular variation can improve estimates on such risk sets. The present paper uses hidden regular variation as a second-order dependence correction layered on top of the original marginal second-order expansion.

2.6 Extreme-value dependence and financial tails

Empirical financial returns are heavy-tailed, clustered, and dependent. The stylized facts literature includes Cont [2001] and the power-law review of Gabaix [2009]. Factor models and portfolio construction are grounded in Fama and French [1993] and Fama and French [2015].

Quantitative risk management and EVT-based VaR/ES methods are treated in McNeil et al. [2015], while simulation for heavy-tailed portfolio VaR is studied by Glasserman et al. [2002]. Backtesting methods include Kupiec [1995], Christoffersen [1998], Engle and Manganelli [2004], and Acerbi and Székely [2014]. Tail-dependence diagnostics build on Ledford and Tawn [1996] and Coles et al. [1999].

2.7 Recent developments and outstanding gaps

Beyond the references above, the post-2020 literature on dependent heavy-tailed simulation has continued to develop along several lines. Efficient importance-sampling schemes for copula and vine-copula models [Cheng et al., 2025] broaden the class of dependence structures for which provably efficient estimators exist. Work on multivariate regular variation under partial mass on cones, on hidden regular variation with multiple asymptotic-independence scales, and on tail-process methods for serially dependent extremes (in the line of Basrak and Segers [2009]) refines the extremal-dependence apparatus. Empirical work on tail-index stability over crisis and non-crisis subperiods continues to motivate rolling-window diagnostics. The present paper integrates these strands at the level of the factor-portfolio model: it states the dependent CdMC identity in a form that specializes to each of latent-shock, block, copula, MRV, and HRV structures, and supplies a real-data protocol that reports the diagnostics needed to choose among them.

Outstanding gaps addressed here. Three gaps motivated the present paper. First, the existing CdMC literature gives BRE results for restricted classes of dependent sums; a unified statement that recovers the independent BRE bound $N^\alpha - 1$ as a special case has been absent. Second, the second-order expansion for finite-portfolio heavy-tailed sums was, prior to the present paper, vulnerable to a specialization failure at $N = 1$ (the local-shift correction term should vanish when there is only one summand). Third, the empirical literature on factor-portfolio tail risk has historically relied on either independent parametric models or fully nonparametric extreme-value methods; an intermediate class that exploits factor structure while admitting principled dependence diagnostics has been missing. Sections 4 to 8 and 10 address these gaps in order.

3 Setup, Notation, and Dependence Classes

3.1 Basic objects

Let $X = (X_1, \dots, X_N)$ be an N -dimensional real-valued loss vector and

$$S_N = \sum_{i=1}^N X_i, \quad M_N = \max_{1 \leq i \leq N} X_i.$$

Positive values are losses. The target rare-event probability is

$$\mu(x) = \mathbb{P}(S_N > x), \quad x \rightarrow \infty.$$

Literature	Representative references	What is established	Role of this paper
Independent heavy-tailed CdMC	Asmussen and Kroese [2006], Chan and Kroese [2011], Hartinger and Kortschak [2009], Asmussen and Kortschak [2015]	Conditional Monte Carlo and efficiency for heavy-tailed sums	Extends i.n.i.d. finite-factor setup, corrected second order, financial pipeline
Dependent heavy-tailed factor models	Djehiche and Svensson [2007], Pourbabaee and Solari [2019]	Factor/idiosyncratic large deviations and simulation ideas	Adds shock-basis constants, estimator hierarchy, real-data diagnostics
Dependent rare-event simulation	Blanchet and Rojas-Nandayapa [2011], Cheng et al. [2025]	Efficient simulation for log-elliptical, copula, and broad dependent structures	Specializes to regularly varying factor portfolios with active sets and spectral controls
MRV and hidden RV	Hult et al. [2005], Maulik and Resnick [2004], Das et al. [2013], Samorodnitsky and Sun [2016]	Vector-tail measures, one large vector step, hidden joint-tail regions	Converts theory into CdMC estimators and financial diagnostics
Financial tail risk	Glasserman et al. [2002], McNeil et al. [2015], Fama and French [2015]	Heavy-tailed VaR/ES modeling and asset-pricing factors	Provides dependent-factor VaR/ES protocol with backtests and attribution

Table 1: Literature map and positioning of the expanded paper. The contribution column states the additional role of the present draft beyond the cited line of work.

Rank	Module		Innovation	Contribution	Feasibility	Recommended status
1	MRV CdMC	spectral/radial	5/5	5/5	3/5	Theoretical centerpiece; best generalization of one big jump
2	Hidden-cone second-order correction		5/5	4.5/5	2/5	High-risk/high-reward; include as template and simulation target
3	Latent/common-shock factors		3.5/5	4.5/5	4.5/5	Best near-term extension and easiest financial interpretation
4	Copula/vine CdMC	conditional	4/5	4/5	3.5/5	Strong applied estimator when conditional kernels are reliable
5	Block-dependent CdMC		3/5	4/5	4/5	Useful empirical bridge and robust implementation option
6	Serial tail-process extension		4/5	3.5/5	2.5/5	Later multi-period extension; not central to this draft

Table 2: Ranking of extension modules by innovation, expected contribution, feasibility, and recommended development status. Scores are qualitative design scores on a five-point scale, not empirical results.

For a vector $a \in \mathbb{R}^d$ and factor return vector $F \in \mathbb{R}^d$, the portfolio factor loss is $L = a^\top F$ after signs have been chosen so that large positive L is a bad portfolio outcome. A residual component may be included as an additional coordinate or block.

A measurable deterministic tie-breaking rule is fixed once and for all. Let $R_i(X) = 1$ denote the event that coordinate i is the selected maximum among $X_1, \dots, X_N$. For example, ties can be broken by smallest index. For each i , write

$$S_{-i} = \sum_{j \neq i} X_j, \quad M_{-i} = \max_{j \neq i} X_j,$$

with $M_{-i} = -\infty$ if $N = 1$. The conditioning threshold for selected maximum decompositions is

$$T_i(x) = \{x - S_{-i}\} \vee M_{-i},$$

with strict or non-strict inequalities adjusted according to the deterministic tie rule. For continuous distributions this distinction is immaterial.

3.2 Regular variation and common reference tails

A positive tail $\bar{G}$ is regularly varying with index $-\alpha$, written $\bar{G} \in \text{RV}_{-\alpha}$, if

$$\frac{\bar{G}(tx)}{\bar{G}(x)} \rightarrow t^{-\alpha}, \quad t > 0.$$

The independent baseline assumes pointwise tail equivalence

$$\mathbb{P}(X_i > x) \sim c_i \bar{G}(x), \quad c_i \geq 0,$$

for a common $\bar{G} \in \text{RV}_{-\alpha}$. The active set is

$$\mathcal{A} = \{i : c_i > 0\}, \quad C = \sum_{i \in \mathcal{A}} c_i.$$

The financial signed-tail constants are obtained by applying this condition to the right tail of each signed contribution. If $X_i = b_i F_i$, then, for a marginal factor with two-sided tail constants p_i^+, p_i^- ,

$$\mathbb{P}(b_i F_i > x) \sim |b_i|^\alpha \{p_i^+ \mathbf{1}(b_i > 0) + p_i^- \mathbf{1}(b_i < 0)\} \bar{G}(x).$$

3.3 Second-order marginal tails

For second-order expansions we use a direct second-order equivalence. For active i , there exists $A(x) \rightarrow 0$, eventually of constant sign, and numbers d_i , such that locally uniformly for u in bounded sets,

$$\frac{\mathbb{P}(X_i > x - u)}{\bar{G}(x)} = c_i \left(1 + \frac{\alpha u}{x}\right) + c_i d_i A(x) + o(A(x) + x^{-1}).$$

This formulation is deliberately local because the second-order term used in the finite-mean expansion shifts the large coordinate by the remaining sum. Inactive or lighter-tail components are assumed negligible at the retained second-order scale when the second-order theorem is invoked.

3.4 Dependence classes

The expanded paper uses five nested or related dependence classes.

Independent coordinates. The X_i 's are independent and have common reference regularly varying tails. This is the original model.

Arbitrary dependence with conditional kernels. The joint law of X is arbitrary, but conditional survival kernels

$$p_i(t; X_{-i}) = \mathbb{P}(X_i > t \mid X_{-i})$$

can be evaluated, approximated, or bounded. This class gives exact unbiasedness identities but not automatic efficiency.

Latent independent shocks. Observed factors are generated by

$$F = BZ, \quad Z_1, \dots, Z_K \text{ independent,}$$

with regularly varying shock tails. A portfolio loss is $a^\top F = q^\top Z$, where $q = B^\top a$. Dependence among observed factors is inherited from the mixing matrix B .

Block dependence. The coordinates are partitioned into blocks $B_1, \dots, B_K$. Blocks are independent; within-block dependence is allowed. If $Y_k = \sum_{i \in B_k} X_i$ has a regularly varying tail, the independent sum theory applies to $Y_1, \dots, Y_K$.

Multivariate regular variation. The full vector satisfies

$$\frac{\mathbb{P}(X/x \in \cdot)}{\bar{H}(x)} \xrightarrow{v} \nu(\cdot)$$

on a cone away from the origin, where $\nu(tA) = t^{-\alpha}\nu(A)$. In polar coordinates $X = R\Theta$, the tail measure decomposes as $\alpha r^{-\alpha-1} dr S(d\theta)$. Axis concentration recovers the independent first-order formula; non-axis spectral mass represents tail dependence.

Hidden regular variation. Ordinary MRV may be axis-concentrated while a slower tail scale exists on pair or block cones. Hidden regular variation is used to quantify joint-tail regions that are invisible under the first MRV limit but relevant for second-order risk.

3.5 Financial loss mapping

Let $r_{p,t}$ be portfolio return and f_t the vector of traded or constructed factors. A rolling factor model is

$$r_{p,t} = \beta_t^\top f_t + \varepsilon_t.$$

The one-day loss is $L_t = -r_{p,t}$. Conditional on rolling estimates, the loss decomposition is

$$L_t = \sum_{j=1}^d X_{j,t} + X_{\varepsilon,t}, \quad X_{j,t} = -\beta_{j,t} f_{j,t}, \quad X_{\varepsilon,t} = -\varepsilon_t.$$

The independent baseline treats the signed factor and residual contributions as independent tail summands. The dependent extensions either transform the factor vector into latent shocks, group factors into blocks, estimate a copula conditional kernel, or estimate the empirical spectral measure of the joint loss-contribution vector.

3.6 Boundary and continuity conventions

For MRV statements, a risk set A is admissible when it is bounded away from the origin and $\nu(\partial A) = 0$. For a linear loss $\ell(z) = a^\top z$, the set $A_\ell = \{z : \ell(z) > 1\}$ is admissible whenever the spectral measure assigns zero mass to $\{\theta : a^\top \theta = 1/r\}$ in the relevant polar normalization, equivalently whenever $\nu\{z : a^\top z = 1\} = 0$. In empirical work we avoid exact-boundary claims by reporting threshold-stability bands.

3.7 Standing assumptions

The following assumptions are invoked throughout the paper. Each main theorem states which subset it requires.

Assumption 3.1 (Common reference regular variation). There exists a survival function $\bar{G}$ with $\bar{G} \in \text{RV}_{-\alpha}$ for some $\alpha > 0$ and constants $c_i \geq 0$ such that $\mathbb{P}(X_i > x) \sim c_i \bar{G}(x)$ as $x \rightarrow \infty$ for every $i = 1, \dots, N$. When the first moment is required we assume $\alpha > 1$.

Assumption 3.2 (Active set non-empty). The active set $\mathcal{A} = \{i : c_i > 0\}$ is non-empty and $C = \sum_{i \in \mathcal{A}} c_i > 0$.

Assumption 3.3 (Tie-breaking rule). A measurable deterministic tie-breaking rule has been fixed; the events $\{R_i(X) = 1\}$ partition the sample space. For continuous joint laws the rule is immaterial.

Assumption 3.4 (Measurability and integrability). The loss vector X is a Borel-measurable random vector on a complete probability space. Conditional survival kernels $p_i(t; X_{-i})$, where invoked, are jointly measurable in their arguments.

Assumption 3.5 (Second-order tail equivalence). For each $i \in \mathcal{A}$ there exists an auxiliary function $A(x) \rightarrow 0$, eventually of constant sign, and a number $d_i \in \mathbb{R}$ such that, locally uniformly in bounded u ,

$$\frac{\mathbb{P}(X_i > x - u)}{\bar{G}(x)} = c_i \left(1 + \frac{\alpha u}{x}\right) + c_i d_i A(x) + o(A(x) + x^{-1}).$$

Inactive components are negligible at the retained second-order scale.

Assumptions 3.1 to 3.4 are the *independent baseline* assumptions; Assumption 3.5 is added for second-order theorems. Dependence-specific assumptions (latent-shock structure, MRV, hidden regular variation) are introduced in Section 5, Section 6, and Section 7 respectively.

4 Independent Baseline

This section records the independent theory inherited from the original manuscript. It is included because every dependent extension is easiest to understand by comparing its tail constant and estimator with the independent benchmark.

4.1 Maximum and sum asymptotics

Assumption 4.1 (Independent pointwise tail equivalence). The random variables $X_1, \dots, X_N$ are independent, $N < \infty$, and there exists $\bar{G} \in \text{RV}_{-\alpha}$, $\alpha > 0$, such that

$$\mathbb{P}(X_i > x) \sim c_i \bar{G}(x), \quad i = 1, \dots, N.$$

At least one c_i is positive.

Theorem 4.2 (Exact leading maximum asymptotic). *Under Assumption 4.1,*

$$\mathbb{P}(M_N > x) = \sum_{i=1}^N \mathbb{P}(X_i > x) + o(\bar{G}(x)) \sim C \bar{G}(x).$$

Proof sketch. Finite inclusion–exclusion: pair intersections are products of marginal tails, hence $O(\bar{G}(x)^2) = o(\bar{G}(x))$, and there are finitely many pairs. The full argument appears in Appendix B. $\square$

The exact constant is C , with no slack factor.

For sums, marginal right-tail equivalence alone is insufficient because large negative components could cancel a large positive component and because many moderate components could combine to cross x . The usual finite-dimensional single-big-jump truncation conditions rule out those pathologies.

Assumption 4.3 (Independent finite-sum single-big-jump conditions). In addition to Assumption 4.1, the left tails are negligible at the right reference scale and there exists $h(x) \uparrow \infty$, $h(x) = o(x)$, such that for each active i ,

$$\mathbb{P}(X_i > x - h(x)) \sim \mathbb{P}(X_i > x),$$

and the probability that two or more truncated components exceed order $h(x)$ is $o(\bar{G}(x))$. Equivalently, the finite collection is in the one-large-jump subexponential regime used for the displayed sum equivalence.

Theorem 4.4 (Independent sum equivalence). *Under Assumption 4.3,*

$$\mathbb{P}(S_N > x) \sim \sum_{i=1}^N \mathbb{P}(X_i > x) \sim C \bar{G}(x).$$

Proof sketch. A truncation argument combined with the pair-exceedance estimate of Theorem 4.2. The full upper and lower bounds appear in Appendix B. $\square$

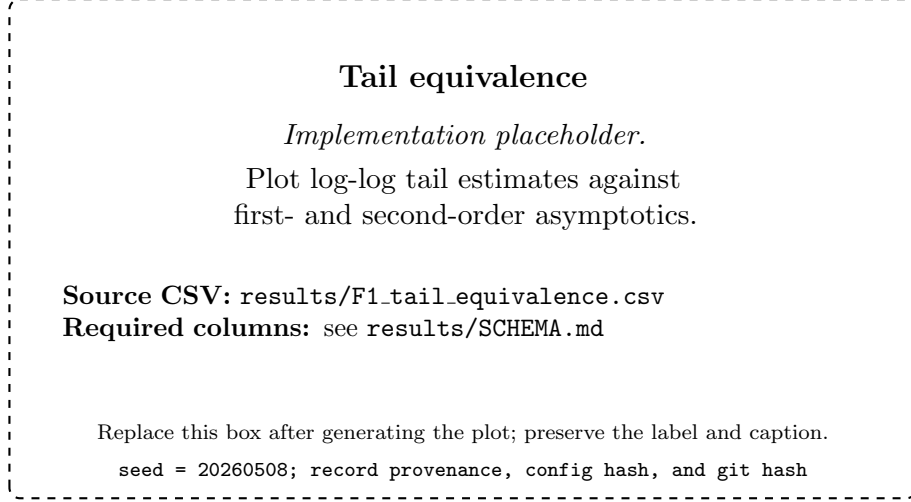


Figure 3: **(PLANNED)**. Planned independent baseline diagnostic. The generated figure will plot $\mathbb{P}(S_N > x)$, $\sum_i \mathbb{P}(X_i > x)$, and the second-order approximation on a log-log scale for fixed N . Source schema: results/F1_tail_equivalence.csv. All curves are placeholders until generated results replace the box.

4.2 Corrected second-order expansion

The second-order term is important for finite thresholds. When X_i is the large component, the first-order shift is created by the remaining sum S_{-i} , not by X_i itself. This yields the leave-one-out mean.

Assumption 4.5 (Second-order tail equivalence and finite means (independent baseline)). The active variables satisfy the local second-order condition Assumption 3.5; $\mathbb{E}|X_i| < \infty$ for all i ; and inactive or lighter-tail summands are negligible at scale $\overline{G}(x)(|A(x)| \vee x^{-1})$.

Theorem 4.6 (Second-order independent expansion). *Under Assumptions 4.3 and 4.5,*

$$\mathbb{P}(S_N > x) = C\overline{G}(x) + \overline{G}(x)A(x) \sum_{i \in \mathcal{A}} c_i d_i + \frac{\alpha \overline{G}(x)}{x} \sum_{i \in \mathcal{A}} c_i \mu_{-i} + o\{\overline{G}(x)(|A(x)| \vee x^{-1})\},$$

where $\mu_{-i} = \sum_{j \neq i} \mathbb{E}X_j$.

Proof sketch. Stratify by the selected-maximum index from Assumption 3.3, localize to $|S_{-i}| \leq h(x)$, and apply the second-order tail equivalence of Assumption 3.5 term by term. The full expansion, including the localization and inactive-component bounds, appears in Appendix B. $\square$

The formula passes the $N = 1$ check because $\mu_{-1} = 0$; a Lomax/Pareto benchmark verifies the cross-term by direct substitution (see Appendix B, verification notes). The constants can be written in closed form and used as a regression test for implementations.

4.3 Independent conditional Monte Carlo

For independent variables define

$$Z^{\text{ind}}(x) = \sum_{i=1}^N \overline{F}_i(T_i(x)), \quad \overline{F}_i(t) = \mathbb{P}(X_i > t).$$

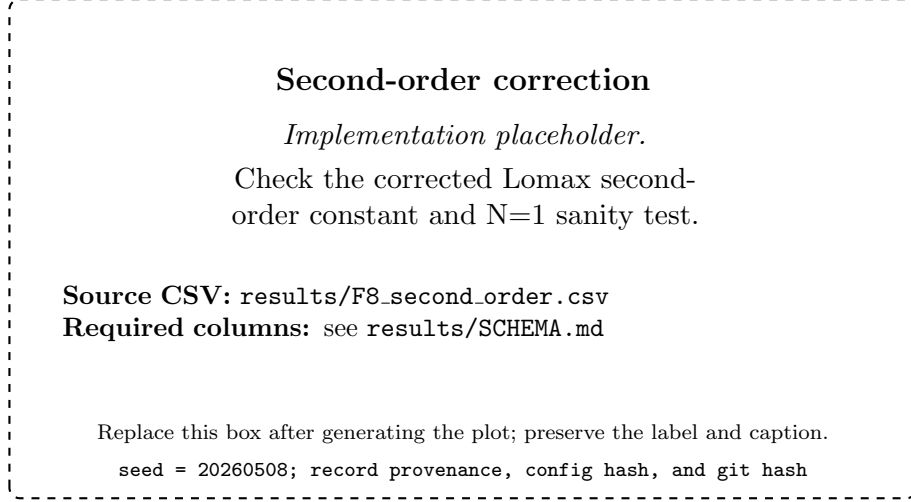


Figure 4: **(PLANNED)**. Planned second-order diagnostic. The generated plot will compare the first-order approximation, the corrected second-order expansion, and a high-precision reference probability across thresholds. The main empirical question is whether the leave-one-out correction improves finite-threshold accuracy. Source schema: **results/F8_second_order.csv**.

Estimator	Tail evals	Status	Guarantee	Main condition
Crude MC ($\hat{\mu}^{\text{CrMC}}$)	0	baseline	No rare-event guarantee	Benchmark only
Summed CdMC (Z^{CdMC})	N	primary	BRE; bound $N^\alpha - 1$	PTE plus single-big-jump conditions
Stratified CdMC (Z^{strat})	1	primary	BRE with contribution weights	Positive exact weights; active-only weights are asymptotic
Closed-form Pareto/Lomax (Z^{ana})	N	implementation	Same variance as summed CdMC	Closed-form marginal survival
Control variate (Z^{VRE})	N	oracle/split	VRE only with safe centering	Local second order and L^2 surrogate accuracy

Table 3: Independent estimator summary from the baseline manuscript. The dependent sections extend the conditioning kernel, shock basis, block basis, and radial basis while keeping this table as the comparison point.

The selected-maximum events partition $\{S_N > x\}$, and independence gives $\mathbb{P}(X_i > T_i \mid X_{-i}) = \bar{F}_i(T_i)$. Thus Z^{ind} is unbiased. Moreover $T_i(x) \geq x/N$ on the relevant scale, so

$$0 \leq Z^{\text{ind}}(x) \leq B(x) := \sum_{i=1}^N \bar{F}_i(x/N).$$

Since $B(x)/\mu(x) \rightarrow N^\alpha$,

$$\limsup_{x \rightarrow \infty} \frac{\text{Var}(Z^{\text{ind}}(x))}{\mu(x)^2} \leq N^\alpha - 1.$$

This is the independent bounded-relative-error constant used throughout the simulation design.

5 Dependent Conditional Monte Carlo and Shock-Basis Extensions

5.1 Exact CdMC identity under arbitrary dependence

The first dependent extension is an identity. It does not require regular variation, independence, or any factor structure; it only requires a regular conditional distribution.

Theorem 5.1 (Arbitrary-dependence selected-maximum CdMC identity). *Let $X \in \mathbb{R}^N$ have any joint distribution admitting regular conditional laws. Fix a deterministic selected-maximum*

tie-breaking rule (Assumption 3.3) and define $T_i(x) = (x - S_{-i}) \vee M_{-i}$, with tie-rule endpoint adjustments on null or discrete boundaries. Then

$$\mathbb{P}(S_N > x) = \sum_{i=1}^N \mathbb{E} [p_i(T_i(x); X_{-i})], \quad p_i(t; X_{-i}) = \mathbb{P}(X_i > t \mid X_{-i}).$$

Consequently

$$Z^{\text{dep}}(x) = \sum_{i=1}^N p_i(T_i(x); X_{-i})$$

is an unbiased estimator of $\mu(x)$ whenever the conditional kernels are evaluated exactly.

Proof sketch. Partition $\{S_N > x\}$ by the selected-maximum rule of Assumption 3.3 into disjoint events $E_i = \{S_N > x, R_i(X) = 1\}$, each of which equals $\{X_i > T_i(x)\}$ conditional on X_{-i} . Take expectations and sum. The full argument and the population-identity / estimator distinction are in Appendix C. $\square$

Remark 5.2 (What independence bought). The independent estimator is the special case $p_i(t; X_{-i}) = \bar{F}_i(t)$. Therefore independence is not the source of unbiasedness. It is the source of a low-dimensional closed-form kernel and of the deterministic envelope $\sum_i \bar{F}_i(x/N)$. Under strong tail dependence, the conditional kernel may become large on high-risk configurations and the independent variance proof no longer applies.

5.2 A sufficient conditional-tail envelope for BRE

A general arbitrary-dependence BRE theorem needs assumptions on conditional survivals. The following condition is intentionally primitive; specialized models below replace it by checkable shock-basis or radial assumptions.

Assumption 5.3 (Conditional-tail envelope). There are nonnegative random variables $K_{i,x} = K_{i,x}(X_{-i})$ and deterministic functions $b_i(x)$ such that, for all $t \geq x/N$,

$$p_i(t; X_{-i}) \leq K_{i,x} b_i(t),$$

where $\sum_i b_i(x/N) = O(\mu(x))$ and

$$\mathbb{E} \left[\left\{ \sum_{i=1}^N K_{i,x} b_i(x/N) \right\}^2 \right] = O(\mu(x)^2).$$

Proposition 5.4 (BRE from a conditional-tail envelope). *Under Assumption 5.3,*

$$\sup_{x \geq x_0} \frac{\text{Var}(Z^{\text{dep}}(x))}{\mu(x)^2} < \infty$$

for all sufficiently large x_0 .

Proof sketch. $T_i(x) \geq x/N$ gives $Z^{\text{dep}}(x) \leq \sum_i K_{i,x} b_i(x/N)$, and the second-moment envelope of Assumption 5.3 controls the variance. The full argument is in Appendix C. $\square$

Remark 5.5 (Identifiability of conditional kernels). The dependent CdMC identity invokes the conditional kernels $p_i(t; X_{-i})$, which depend on the joint law of X . In practice these kernels must either be derived in closed form from a parametric model (e.g. a copula or latent-shock model) or estimated from data. Identifiability of the kernel from observed data is not automatic: under strong tail dependence the support of X_{-i} relevant for the tail integral may be sparsely populated in finite samples. The latent-shock, block, copula, MRV, and HRV specializations developed below exist precisely to choose a more identifiable conditioning object than the raw coordinatewise kernel.

This proposition is useful as a diagnostic but not as a final model. It shows why coordinatewise dependent CdMC can work under weak extremal dependence and why it can fail under common shocks. If X_j is already extreme, the conditional probability that X_i is also extreme may be much larger than $\bar{F}_i(t)$. In that regime the estimator should condition on the common shock, block, or radial component instead.

5.3 Copula and vine implementation of the exact kernel

If $U_i = F_i(X_i)$ and U has copula C , then the exact kernel is

$$p_i(t; X_{-i}) = 1 - C_{i|-i}\{F_i(t) \mid U_{-i}\}.$$

For elliptical, Archimedean, and vine copulas this conditional distribution can be computed by standard conditional copula formulas or by vine h-functions. The planned copula estimator is therefore

$$Z^{\text{cop}}(x) = \sum_{i=1}^N [1 - C_{i|-i}\{F_i(T_i(x)) \mid U_{-i}\}].$$

The simulation section tests this estimator in Gaussian, t , Clayton, Gumbel, and vine designs. The real-data section uses it only after reporting marginal tail and copula-tail diagnostics, because misspecified copulas can be precisely wrong in the region that matters.

5.4 Latent independent shocks

Suppose observed factors are dependent because they are linear combinations of independent shocks:

$$F = BZ, \quad Z_1, \dots, Z_K \text{ independent.}$$

For portfolio exposure a , define $q = B^\top a$, so $L = a^\top F = q^\top Z$. Let Z_k have right and left tail constants p_k^+, p_k^- relative to $\bar{G}$. The active shock constant is

$$c_k(q_k) = |q_k|^\alpha \{p_k^+ \mathbf{1}(q_k > 0) + p_k^- \mathbf{1}(q_k < 0)\}.$$

Theorem 5.6 (Latent-shock tail constant). *Assume the shocks $Z_1, \dots, Z_K$ are independent and satisfy the independent single-big-jump assumptions after signing by q_k . Then*

$$\mathbb{P}(q^\top Z > x) \sim \bar{G}(x) \sum_{k=1}^K c_k(q_k).$$

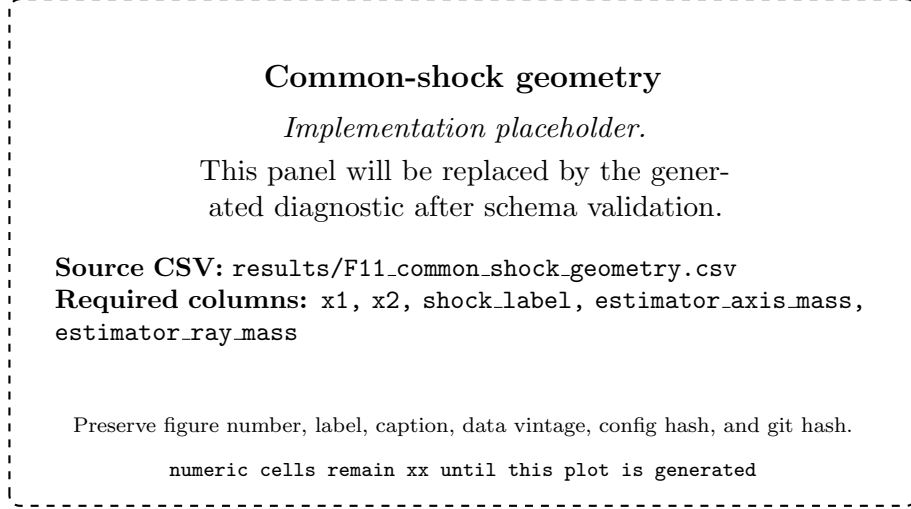


Figure 5: **(PLANNED)**. Common-shock geometry. The final figure will show simulated observations from $X_i = b_i Z_0 + E_i$, highlighting that joint extremes align with the common-shock ray. The independent estimator assigns mass to coordinate axes, while the latent-shock estimator assigns mass to the ray. Source schema: results/F11.common.shock.geometry.csv.

Moreover the independent CdMC estimator applied to the shock contributions $q_k Z_k$ is unbiased and has the independent BRE bound with K in place of N , or with the number of active shocks after active-only stratification.

Proof sketch. Apply Theorem 4.4 to the independent signed shock contributions $Y_k = q_k Z_k$. The full argument is in Appendix C. $\square$

The theorem is elementary but consequential: the unit of tail attribution is a latent shock, not necessarily an observed factor.

Example 5.7 (Common shock). Let $X_i = b_i Z_0 + E_i$, where $Z_0, E_1, \dots, E_N$ are independent and regularly varying with common index. Then

$$S_N = \left(\sum_{i=1}^N b_i \right) Z_0 + \sum_{i=1}^N E_i.$$

The common-shock contribution to the leading right tail is proportional to $(\sum_i b_i)_+^\alpha$. Treating the observed X_i 's as independent would instead create a term proportional to $\sum_i (b_i)_+^\alpha$, which can be severely wrong when the exposures have the same sign.

5.5 Block-dependent aggregation

Let $\{1, \dots, N\} = B_1 \cup \dots \cup B_K$ be a partition. Define block sums $Y_k = \sum_{i \in B_k} X_i$. Assume the blocks are independent, while dependence inside each block is arbitrary.

Theorem 5.8 (Independent-block reduction). *If $Y_1, \dots, Y_K$ are independent and*

$$\mathbb{P}(Y_k > x) \sim d_k \overline{G}(x),$$

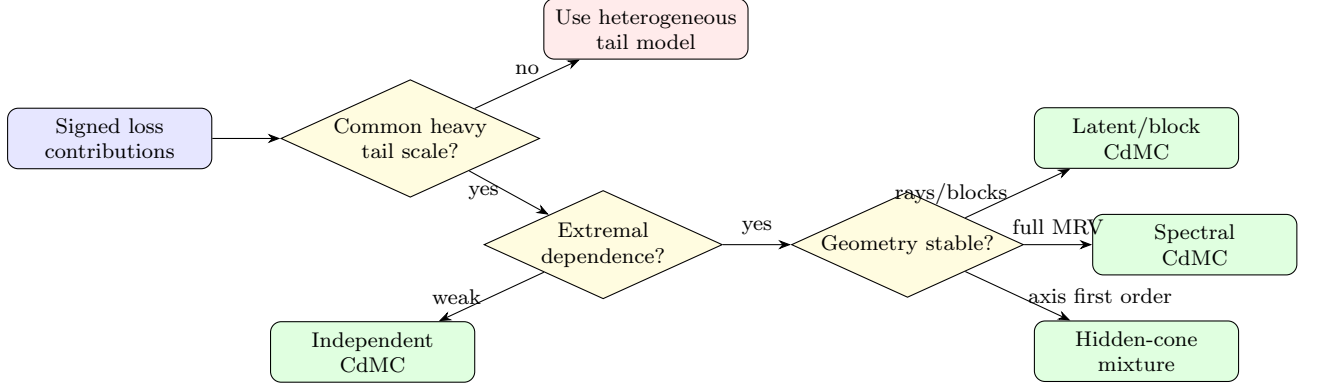


Figure 6: Estimator decision tree for dependent factor risk. The tree is used both in simulation and in the real-data analysis. It is a model-selection protocol, not a claim that one estimator dominates universally.

and the block sums satisfy the finite single-big-jump conditions, then

$$\mathbb{P}(S_N > x) = \mathbb{P}\left(\sum_{k=1}^K Y_k > x\right) \sim \sum_{k=1}^K \mathbb{P}(Y_k > x) \sim \left(\sum_{k=1}^K d_k\right) \bar{G}(x).$$

Proof sketch. Apply Theorem 4.4 to the independent block sums. The full argument is in Appendix C. $\square$

In financial applications, blocks may be sectors, regions, liquidity buckets, volatility regimes, or factors grouped by empirical tail dependence. The block constant d_k can be estimated by within-block MRV, a common-shock model, a copula estimator, or high-precision simulation.

Remark 5.9 (Numerical stability of conditional-kernel evaluation). When the conditional kernel $p_i(t; X_{-i})$ is computed from a closed-form survival expression that is under-flow prone at large t (for example, Gaussian or Student- t conditional tails at the 99.9% quantile), the kernel should be evaluated on log-scale: compute $\log p_i$ via Mills'-ratio or asymptotic series and exponentiate at the last step. Sum-of-kernel estimators $Z^{\text{dep}}(x) = \sum_i p_i(T_i; X_{-i})$ then aggregate the exponentiated quantities; if any single kernel under-flows to zero, the log-sum-exp identity may be used at the outer level to retain accuracy in the overall estimator.

5.6 Estimator decision point

The practical decision is not whether dependence exists; it does. The decision is whether dependence changes the first-order constant materially at the thresholds of interest. The paper therefore uses the workflow in Figure 6: start with marginal tails, test tail dependence and spectral concentration, choose independent/latent/block/copula or MRV estimator, then backtest VaR and ES out of sample.

6 Multivariate Regular Variation and Spectral CdMC

6.1 MRV tail measure

The most general first-order dependence extension is multivariate regular variation. We use the standard definition.

Definition 6.1 (Multivariate regular variation). Let $X \in \mathbb{R}^N$ and $\mathcal{E} \subset \mathbb{R}^N \setminus \{0\}$ a cone. We say that X is multivariate regularly varying (MRV) on $\mathcal{E}$ with limit measure ν and tail scale $\overline{H} \in \text{RV}_{-\alpha}$, written $X \in \text{MRV}(\alpha, \overline{H}, \nu, \mathcal{E})$, if ν is a non-zero Radon measure on $\mathcal{E}$ satisfying $\nu(tA) = t^{-\alpha}\nu(A)$ for all $t > 0$ and Borel $A \subset \mathcal{E}$ bounded away from the origin, and

$$\frac{\mathbb{P}(X/x \in \cdot)}{\overline{H}(x)} \xrightarrow{v} \nu(\cdot) \quad \text{vaguely on } \mathcal{E},$$

i.e. $\mathbb{P}(X/x \in A)/\overline{H}(x) \rightarrow \nu(A)$ for every Borel $A \subset \mathcal{E}$ bounded away from the origin with $\nu(\partial A) = 0$.

In polar coordinates $X = R\Theta$ the homogeneity of ν factorizes the limit measure as $\nu(\text{d}r, \text{d}\theta) = \alpha r^{-\alpha-1} \text{d}r S(\text{d}\theta)$, where the angular measure S on the unit sphere or simplex encodes the directional concentration of extreme vectors. The spectral measure is identifiable up to the choice of radial normalization.

Definition 6.2 (Spectral measure). The angular (spectral) measure S associated to an MRV law is the unique finite Radon measure on the unit sphere $\mathbb{S}^{N-1}$ (or on the unit simplex when the cone is restricted to the positive orthant) satisfying the polar decomposition above for the chosen radial normalization.

For a linear loss $\ell(z) = a^\top z$, define

$$A_\ell = \{z : \ell(z) > 1\}.$$

Assumption 6.3 (MRV loss set is admissible). The loss vector X satisfies $X \in \text{MRV}(\alpha, \overline{H}, \nu, \mathcal{E})$ for some $\alpha > 0$, tail scale $\overline{H} \in \text{RV}_{-\alpha}$, limit measure ν , and cone $\mathcal{E} \supseteq A_\ell$. The set $A_\ell = \{z : \ell(z) > 1\}$ is bounded away from the origin and $\nu(\partial A_\ell) = 0$.

Theorem 6.4 (MRV linear-risk asymptotic). *Under Assumption 6.3,*

$$\mathbb{P}\{\ell(X) > x\} \sim \overline{H}(x)\nu(A_\ell).$$

If ν has polar representation

$$\nu(\text{d}r, \text{d}\theta) = \alpha r^{-\alpha-1} \text{d}r S(\text{d}\theta),$$

with $r > 0$ and angular measure S , then

$$\nu(A_\ell) = \int (\ell(\theta)_+)^{\alpha} S(\text{d}\theta),$$

under the chosen radial normalization.

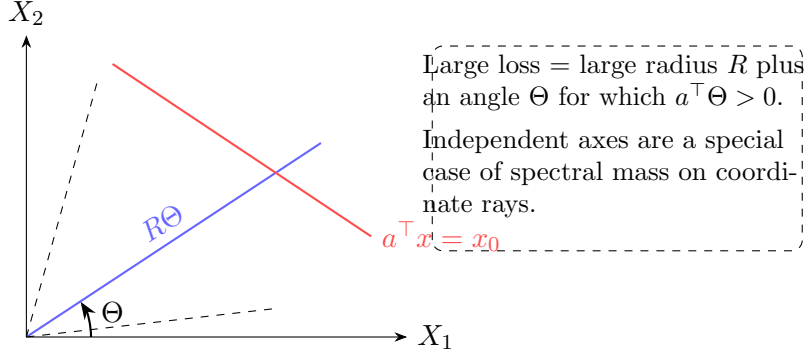


Figure 7: Radial-angular decomposition of a dependent heavy-tailed factor vector. The radial variable supplies the large magnitude; the spectral direction determines which factors move together and how much the portfolio loss functional amplifies the event.

Proof sketch. Apply Definition 6.1 to A_ℓ , then change to polar coordinates and integrate radially. The full argument is in Appendix C. $\square$

This theorem contains the independent leading constant as a special case. If S is concentrated on coordinate axes e_i , with masses proportional to c_i , and $\ell(z) = \sum_i z_i$, then the integral collapses to $\sum_i c_i$. If S has mass away from the axes, joint extreme directions contribute at first order.

6.2 Spectral estimator

Suppose we can simulate or resample from an angular distribution $S/S(\Theta)$ and evaluate the conditional radial survival. For $X = R\Theta$,

$$\ell(X) > x \iff R > x/\ell(\Theta),$$

when $\ell(\Theta) > 0$. The radial conditional estimator is

$$Z^{\text{spec}}(x) = \mathbb{P}\{R > x/\ell(\Theta) \mid \Theta\} \mathbf{1}\{\ell(\Theta) > 0\}.$$

Proposition 6.5 (Exact radial conditional estimator). *If the joint law admits a radial-angular representation with regular conditional radial survival $\bar{F}_R(r \mid \Theta)$, then*

$$Z^{\text{spec}}(x) = \bar{F}_R\{x/\ell(\Theta) \mid \Theta\} \mathbf{1}\{\ell(\Theta) > 0\}$$

is unbiased for $\mathbb{P}\{\ell(X) > x\}$.

In an exact Pareto radial model with $\mathbb{P}(R > r \mid \Theta) = r^{-\alpha}$ above a threshold and with $x/\ell(\Theta)$ above that threshold,

$$Z^{\text{spec}}(x) = x^{-\alpha} (\ell(\Theta)_+)^{\alpha}.$$

The relative variance is then independent of x and equals

$$\frac{\mathbb{E}[(\ell(\Theta)_+)^{2\alpha}]}{\mathbb{E}[(\ell(\Theta)_+)^{\alpha}]^2} - 1,$$

provided the numerator is finite.

Proposition 6.6 (Spectral BRE). *Assume the conditional radial survival satisfies, uniformly over the angular support used by the estimator,*

$$\overline{F}_R\{x/y \mid \Theta\} \leq K\overline{H}(x)y^\alpha, \quad y = \ell(\Theta)_+,$$

for large x , and $0 < \mathbb{E}[y^\alpha] \leq \mathbb{E}[y^{2\alpha}] < \infty$. Then the radial estimator has bounded relative error.

6.3 Estimating the spectral measure

The empirical spectral measure is estimated from exceedances of a radial norm $\|X_t\|$ above a high threshold u . For selected exceedances,

$$\widehat{\Theta}_t = X_t / \|X_t\|.$$

The empirical spectral estimator of the first-order constant is

$$\widehat{C}_\ell(u) = \frac{1}{k} \sum_{t: \|X_t\| > u} (\ell(\widehat{\Theta}_t)_+)^{\widehat{\alpha}},$$

with threshold-stability and bootstrap bands. In the real-data section this is reported for the FF3, FF5, FF5+momentum, industry-portfolio, and security-level contribution vectors.

Remark 6.7 (Identifiability of the spectral measure). The angular measure S is identifiable only up to the chosen radial normalization (the norm $\|\cdot\|$ used to define $\Theta = X/\|X\|$ and the radial threshold u). Different choices of norm produce different spectral measures with the same first-order linear-risk constant $\nu(A_\ell)$ provided the boundary continuity of Assumption 6.3 holds. The empirical spectral estimator $\widehat{C}_\ell(u)$ is consistent for $\nu(A_\ell)$ as $k \rightarrow \infty$, $u \rightarrow \infty$, and $k/n \rightarrow 0$ at appropriate rates; convergence rates require additional second-order regularity (see, e.g., ?). In empirical work we report $\widehat{C}_\ell(u)$ over a threshold grid and flag instability when no plateau is observed.

Definition 6.8 (Tail equivalence in the MRV sense). Two non-negative random vectors X, Y on a cone $\mathcal{E}$ are MRV-tail-equivalent if there exists a common tail scale $\overline{H} \in \text{RV}_{-\alpha}$ and a non-zero Radon measure ν on $\mathcal{E}$ such that both $X \in \text{MRV}(\alpha, \overline{H}, \nu, \mathcal{E})$ and $Y \in \text{MRV}(\alpha, \overline{H}, \nu, \mathcal{E})$ in the sense of Definition 6.1. When X has N scalar coordinates and the limit measure ν is axis-concentrated with marginal weights $c_1, \dots, c_N$, MRV-tail-equivalence reduces to the univariate tail equivalence $\mathbb{P}(X_i > x) \sim c_i \overline{G}(x)$ of Assumption 3.1, recovering the independent baseline.

6.4 Comparison with dependent-sum simulation

MRV gives a first-order constant and a natural estimator when the vector itself is the extreme object. It is complementary to the dependent-sum simulation literature, which often constructs importance samplers by conditioning on one large component or by transforming dependent structures such as log-elliptical vectors. The spectral estimator is most attractive when empirical angular data show stable non-axis mass and when a radial tail model is credible. If angular mass is concentrated on a small number of rays, a latent-shock model may be more parsimonious. If

angular mass is concentrated on coordinate axes but pair cones are non-negligible at a slower scale, hidden regular variation is the next layer.

7 Hidden Regular Variation and Dependent Second-Order Terms

7.1 Why hidden cones matter

Ordinary MRV may report asymptotic independence: the limiting spectral measure is concentrated on coordinate axes. In that case the first-order tail constant for S_N can look identical to the independent constant. This does not mean that dependence is irrelevant at realistic thresholds. Joint-tail mass can live on smaller cones at a slower scale. Hidden regular variation formalizes that phenomenon and is especially relevant for financial data, where two factors may rarely be extreme together but still co-crash often enough to affect 99.5% or 99.9% risk estimates.

7.2 Cone decomposition and hidden regular variation

For simplicity consider the positive loss cone. Let

$$\mathcal{E}_1 = [0, \infty)^N \setminus \{0\}, \quad \mathcal{E}_2 = [0, \infty)^N \setminus \bigcup_{i=1}^N \{te_i : t > 0\}.$$

The first cone captures all extremes; the second removes the coordinate axes and captures joint extremes.

Definition 7.1 (Hidden regular variation). Let X be a non-negative random vector with $X \in \text{MRV}(\alpha, \bar{G}, \nu_1, \mathcal{E}_1)$ and suppose ν_1 is concentrated on the coordinate axes. We say that X exhibits hidden regular variation on $\mathcal{E}_2$ with hidden tail scale $\bar{H}_2$ and hidden limit measure ν_2 , written $X \in \text{HRV}(\alpha_2, \bar{H}_2, \nu_2, \mathcal{E}_2)$, if $\bar{H}_2 \in \text{RV}_{-\alpha_2}$ with $\alpha_2 \geq \alpha$ and

$$\frac{\mathbb{P}(X/x \in \cdot)}{\bar{H}_2(x)} \xrightarrow{v} \nu_2(\cdot) \quad \text{vaguely on } \mathcal{E}_2,$$

with ν_2 non-zero, homogeneous of order $-\alpha_2$, and supported on $\mathcal{E}_2$. When $\alpha_2 > \alpha$, $\bar{H}_2(x) = o(\bar{G}(x))$ but $\bar{H}_2$ may exceed the marginal second-order scale $\bar{G}(x)|A(x)|$ or the mean-shift scale $\bar{G}(x)/x$.

Hidden regular variation in the sense of Definition 7.1 originates with [Ledford and Tawn \[1996\]](#), [Resnick \[2002\]](#), and [Maulik and Resnick \[2005\]](#); [Das et al. \[2013\]](#) emphasize the empirical consequences for risk regions away from the axes. The combination of an axis-supported first-order ν_1 and a hidden ν_2 on $\mathcal{E}_2$ formalizes the observation that two factors may rarely be jointly extreme but co-crash often enough to affect 99.5% and 99.9% risk estimates.

Assumption 7.2 (Ordered hidden-cone scale). The first-order measure ν_1 is axis-supported (i.e. assigns zero mass to $\mathcal{E}_2$); X exhibits hidden regular variation on $\mathcal{E}_2$ with scale $\bar{H}_2$ and limit measure ν_2 in the sense of Definition 7.1; and the risk-set boundaries for the sum functional are null under both ν_1 and ν_2 . Let

$$r(x) = \max\{\bar{H}_2(x), \bar{G}(x)|A(x)|, \bar{G}(x)/x\}.$$

The remainder from triple and higher hidden cones not explicitly modeled is $o(r(x))$, or those cones are absorbed into ν_2 as block cones.

Theorem 7.3 (Hidden-cone second-order expansion). *Under Assumptions 3.5, 4.3 and 7.2 with axis first-order measure and hidden cone measure ν_2 ,*

$$\mathbb{P}(S_N > x) = C_{\text{axis}}\overline{G}(x) + D_{\text{hid}}\overline{H}_2(x) + \overline{G}(x)A(x)D_{\text{marg}} + \frac{\alpha\overline{G}(x)}{x}D_{\text{mean}} + o(r(x)),$$

where

$$D_{\text{hid}} = \nu_2\{z : \sum_i z_i > 1\}, \quad D_{\text{marg}} = \sum_{i \in \mathcal{A}} c_i d_i, \quad D_{\text{mean}} = \sum_{i \in \mathcal{A}} c_i \mu_{-i}.$$

Proof sketch. Split the risk set into an axis neighbourhood and the hidden-cone region $\mathcal{E}_2$. The axis region contributes the marginal second-order and mean-shift terms by the localization argument of Theorem 4.6; the hidden region contributes $\overline{H}_2(x)\nu_2(A_S \cap \mathcal{E}_2)$ by Definition 7.1; the overlap between the two regions is null by construction. The full argument is in Appendix D. $\square$

This theorem is a template result: the proof is rigorous once the cone scales and boundary assumptions are verified for a concrete model. The planned simulation designs include Gaussian copula asymptotic independence, selected Archimedean copulas, and mixture models that create controllable hidden-cone mass.

7.3 Mixture estimator for axis and hidden cones

A practical estimator should allocate effort to the axis event and to hidden joint-tail events. Let $Z_{\text{axis}}(x)$ estimate the axis component and $Z_{\text{hid}}(x)$ estimate the hidden-cone component. With mixture probability π_x , define

$$Z^{\text{mix}}(x) = \frac{\mathbf{1}\{I = 0\}}{\pi_x} Z_{\text{axis}}(x) + \frac{\mathbf{1}\{I = 1\}}{1 - \pi_x} Z_{\text{hid}}(x),$$

where $I \sim \text{Bernoulli}(1 - \pi_x)$ is independent of the simulation draws. Choosing π_x proportional to preliminary estimates of the axis and hidden components gives a stratified estimator whose variance does not waste samples on a cone that is negligible at the current threshold.

7.4 Empirical diagnostics for hidden variation

The real-data analysis reports three hidden-cone diagnostics. First, for each pair (i, j) , it estimates the Ledford–Tawn residual tail-dependence slope from

$$\mathbb{P}(U_i > 1 - s, U_j > 1 - s) \approx L(s)s^{1/\eta_{ij}},$$

where U_i are marginal ranks. Second, it estimates pair-cone exceedance rates conditional on $\|X\| > u$. Third, it compares pair-cone risk contributions with the independent second-order correction at the VaR/ES thresholds. The results populate Table 12 and Figure 13.

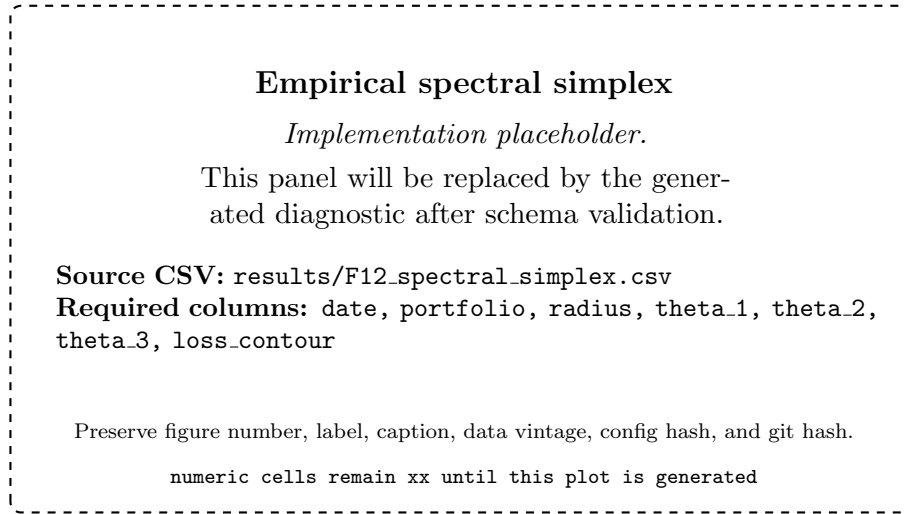


Figure 8: Empirical spectral-measure placeholder. The generated figure will plot angular exceedances for selected factor-contribution vectors on a simplex or projected sphere, with axis mass, common-shock rays, and portfolio-loss contours. Source schema: `results/F12_spectral_simplex.csv`.

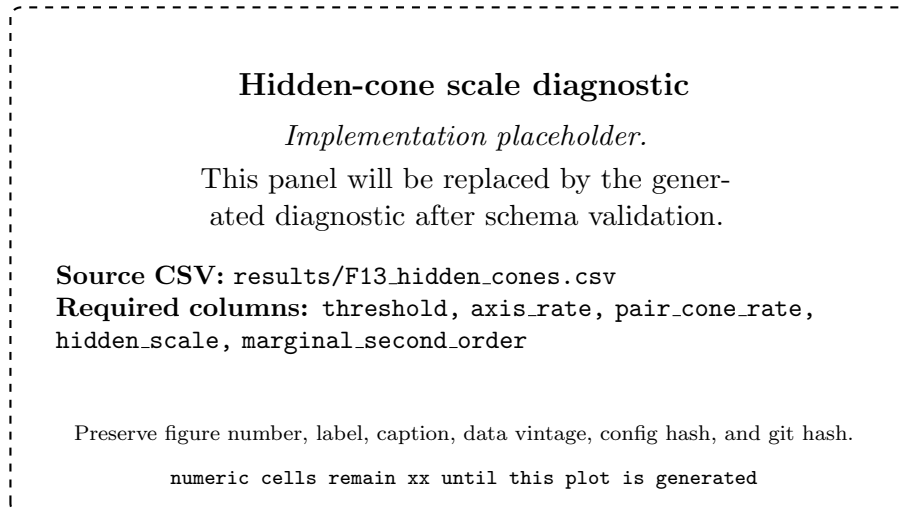


Figure 9: **(PLANNED)**. Hidden-cone diagnostic. The final figure will compare axis exceedances, pair-cone exceedances, and block-cone exceedances across thresholds. It will report whether the hidden scale is larger than the marginal second-order and mean-shift scales over the empirical range. Source schema: `results/F13_hidden_cones.csv`.

8 Estimator Families and Efficiency Criteria

8.1 Estimator menu

The expanded paper compares six estimator families.

Crude Monte Carlo. Simulate X and average $\mathbf{1}\{S_N > x\}$. This is the baseline for sample-size and work-normalized comparisons.

Independent summed CdMC. Use $\sum_i \bar{F}_i(T_i)$. This is exact only under independence but remains a useful misspecification benchmark.

Dependent kernel CdMC. Use $\sum_i \mathbb{P}(X_i > T_i \mid X_{-i})$. This is exact under arbitrary dependence if conditional kernels are correct.

Latent-shock CdMC. Transform to independent shocks and apply independent CdMC to signed shock contributions.

Block CdMC. Estimate or model block-tail probabilities and apply CdMC to independent block sums.

Spectral/radial CdMC. Condition on angular direction and evaluate radial survival. This is the preferred estimator when MRV with non-axis spectral mass is empirically supported.

8.2 Efficiency definitions

We collect the efficiency notions used throughout the paper. Throughout, $Z(x)$ denotes a generic unbiased estimator of $\mu(x)$, n the replication count, and $\text{cost}(Z)$ the average wall-clock cost per replicate (or equivalently, the count of tail-function evaluations per replicate when wall-clock work is dominated by tail evaluations).

Definition 8.1 (Bounded relative error, vanishing relative error, log-efficiency). An unbiased estimator $Z(x)$ for $\mu(x)$:

- has *bounded relative error* (BRE) if $\limsup_{x \rightarrow \infty} \text{Var } Z(x) / \mu(x)^2 < \infty$;
- has *vanishing relative error* (VRE) if $\text{Var } Z(x) / \mu(x)^2 \rightarrow 0$;
- is *logarithmically efficient* if $\liminf_{x \rightarrow \infty} \log \mathbb{E} Z(x)^2 / \log \mu(x)^2 \geq 1$.

Definition 8.2 (Work, work-normalized variance, efficiency frontier). The *work* of $Z(x)$ at threshold x is the mean cost per replicate $\text{cost}(Z; x) \in (0, \infty)$. The *work-normalized variance* is

$$W(Z; x) = \text{Var } Z(x) \cdot \text{cost}(Z; x),$$

the *work-normalized relative error* (WNRE) is

$$\text{WNRE}(Z; x) = \sqrt{W(Z; x)} / \mu(x),$$

Estimator	Conditioning object	Required model input	Exactness condition	Planned diagnostic
Independent CdMC	Observed coordinate	Marginal functions	Observed contributions independent	Bias under dependence; BRE replication
Dependent kernel CdMC	Observed coordinate given others	Conditional kernels	Conditional kernels correctly specified	Kernel calibration and relative error
Latent-shock CdMC	Independent shock	Shock mixing matrix and shock tails	Shock representation correctly specified	Shock attribution and common-shock bias reduction
Block CdMC	Block sum	Block partition and block-tail model	Blocks independent or weakly coupled	Block stability and work-normalized error
Copula CdMC	Conditional copula coordinate	Margins plus conditional copula	Copula and margins correctly specified	Tail-dependence fit and backtest performance
Spectral CdMC	Angular direction	Spectral sample and radial tail	MRV angular model valid	Spectral mass stability and radial tail fit
Hidden-cone mixture	Axis/hidden stratum	Hidden-cone and sampler	Hidden scale estimated reliably	Hidden term versus second-order scales

Table 4: Dependent estimator summary. The table specifies the conditioning object, required model input, exactness condition, and planned efficiency diagnostic for each estimator family.

and the *efficiency ratio* of Z_1 relative to Z_2 is $\text{eff}(Z_1, Z_2; x) = W(Z_2; x)/W(Z_1; x)$. The *efficiency frontier* over a class of estimators $\mathcal{Z}$ at threshold x is the lower convex boundary of the set $\{(\text{cost}(Z; x), \text{Var } Z(x)) : Z \in \mathcal{Z}\}$.

In the empirical pipeline $\widehat{c}_{\text{pu}}$ denotes the sample estimate of $\text{cost}(Z; x)$ and $\widehat{\text{WNRE}}(Z; x)$ replaces the population quantities by their sample analogues.

8.3 Efficiency results for the principal estimators

The independent and dependent CdMC efficiency results proved in Appendices B and C are summarized here as formal propositions for ease of reference.

Proposition 8.3 (BRE of independent CdMC). *Under Assumptions 4.1 and 4.3, the independent summed CdMC estimator $Z^{\text{ind}}(x)$ is unbiased for $\mu(x)$ and satisfies $\limsup_{x \rightarrow \infty} \text{Var } Z^{\text{ind}}(x)/\mu(x)^2 \leq N^\alpha - 1$. The bound is scale-invariant under common rescaling of the X_i .*

Proof sketch. Established in Appendix B via the deterministic envelope $Z^{\text{ind}}(x) \leq \sum_i \bar{F}_i(x/N)$ and the regular-variation limit $\sum_i \bar{F}_i(x/N)/\mu(x) \rightarrow N^\alpha$. $\square$

Proposition 8.4 (BRE of dependent CdMC under a conditional-tail envelope). *Under Assumption 5.3, the dependent CdMC estimator $Z^{\text{dep}}(x)$ is unbiased for $\mu(x)$ and satisfies $\limsup_{x \rightarrow \infty} \text{Var } Z^{\text{dep}}(x)/\mu(x)^2 < \infty$.*

Proof sketch. This is Proposition 5.4; see Appendix C. $\square$

Proposition 8.5 (BRE of latent-shock CdMC). *If $F = BZ$ with $Z_1, \dots, Z_K$ independent and regularly varying with common index, and Assumption 4.3 holds for the signed shock contributions $q_k Z_k$, then the latent-shock CdMC estimator is unbiased and satisfies the independent BRE bound $K^\alpha - 1$ in place of $N^\alpha - 1$.*

Proof sketch. This is Theorem 5.6 combined with the BRE bound for the independent CdMC applied to the shock vector. The full argument is in Appendix C. $\square$

Proposition 8.6 (BRE of spectral CdMC). *Under Assumption 6.3 with $\mathbb{E}[\ell(\Theta)_+^{2\alpha}] < \infty$ and the envelope condition of Proposition 6.6, the spectral/radial CdMC estimator $Z^{\text{spec}}(x)$ is unbiased and has bounded relative error.*

Proof sketch. Proposition 6.6; full argument in Appendix C. $\square$

8.4 Control variates and the VRE estimator

The independent second-order approximation and the MRV spectral approximation both produce analytic or semi-analytic controls. If $Y(x)$ has known mean $m_Y(x)$, then

$$Z^{\text{cv}}(x) = Z(x) - \gamma\{Y(x) - m_Y(x)\}$$

is unbiased. The oracle coefficient is

$$\gamma^*(x) = \frac{\text{Cov}(Z(x), Y(x))}{\text{Var}(Y(x))}.$$

In practice γ and sometimes m_Y are estimated on a pilot split.

Proposition 8.7 (Exact and asymptotic VRE for the centered control-variate estimator). *Let $Y(x)$ be an L^2 surrogate for $Z(x)$ with $\text{Var } Z(x), \text{Var } Y(x) > 0$ and let $\rho(x) = \text{Corr}(Z(x), Y(x))$. Then:*

1. (Exact VRE with oracle centering.) *If $m_Y(x)$ is known exactly and the oracle coefficient $\gamma^*(x)$ is used, then Z^{cv} is unbiased and*

$$\text{Var } Z^{\text{cv}}(x) = (1 - \rho(x)^2) \text{Var } Z(x).$$

If $\rho(x) \rightarrow 1$ as $x \rightarrow \infty$, then $\text{Var } Z^{\text{cv}}(x)/\mu(x)^2 \rightarrow 0$ provided $Z(x)$ has BRE; that is, Z^{cv} has VRE.

2. (Asymptotic VRE with sample-split centering.) *If $m_Y(x)$ is estimated on a pilot split of size n_0 and γ on a second split, then Z^{cv} is unbiased as long as the splits are independent of the production sample, but its variance picks up an additional term of order $1/n_0$ from the centering estimate. Under the same conditions as part (1) and provided $n_0 \rightarrow \infty$ at any rate with x , the work-normalized variance ratio $W(Z^{\text{cv}}; x)/\text{Var } Z(x) \rightarrow 0$ along the same threshold sequence.*

Proof sketch. Part (1) is the standard identity for an oracle-centered control variate; the VRE implication follows because $1 - \rho(x)^2 \rightarrow 0$ kills the surviving BRE constant. Part (2) is the standard sample-splitting argument with an independent pilot. Details mirror the classical control-variate analysis in Glasserman [2004]; see Appendix C for the bookkeeping. $\square$

The distinction in Proposition 8.7 matters for empirical claims: exact centering yields VRE, while sample-split centering yields VRE only along the joint limit $x \rightarrow \infty, n_0 \rightarrow \infty$. The paper reports both variants.

8.5 Finite-sample concentration

If an estimator is bounded by $0 \leq Z(x) \leq B_Z(x)$, Hoeffding gives

$$\mathbb{P}(|\bar{Z}_n - \mu(x)| > \varepsilon \mu(x)) \leq 2 \exp \left\{ -\frac{2n\varepsilon^2 \mu(x)^2}{B_Z(x)^2} \right\}.$$

A tighter Bernstein-style bound that exploits the small variance of CdMC estimators is the following.

Theorem 8.8 (Bernstein-type confidence interval for bounded CdMC estimators). *Let $Z(x)$ be an unbiased estimator of $\mu(x)$ with $0 \leq Z(x) \leq B_Z(x)$ and $\text{Var } Z(x) \leq \sigma^2(x)$. Let $\bar{Z}_n$ denote the sample mean of n i.i.d. replicates. Then for all $\varepsilon > 0$,*

$$\mathbb{P}(|\bar{Z}_n - \mu(x)| > \varepsilon) \leq 2 \exp \left\{ -\frac{n\varepsilon^2}{2\sigma^2(x) + \frac{2}{3}B_Z(x)\varepsilon} \right\}.$$

In particular, taking $\varepsilon = \delta\mu(x)$ and using the BRE bound $\sigma^2(x)/\mu(x)^2 \leq V$ and the envelope

ratio $B_Z(x)/\mu(x) \leq M$ gives

$$\mathbb{P}(|\bar{Z}_n - \mu(x)| > \delta\mu(x)) \leq 2 \exp\left\{-\frac{n\delta^2}{2V + \frac{2}{3}M\delta}\right\}.$$

Proof sketch. This is Bernstein’s inequality applied to the i.i.d. sequence $Z_1(x), \dots, Z_n(x)$, with the variance and uniform bound substituted in. The substitution to relative-error form uses the BRE bound on σ^2/μ^2 and the deterministic envelope ratio B_Z/μ ; for the independent CdMC, $M = N^\alpha$ and $V = N^\alpha - 1$ by Proposition 8.3. $\square$

The simulation study reports Bernstein intervals from Theorem 8.8, Hoeffding intervals when only a deterministic range bound is available, and self-normalized confidence sequences when heavy-tailed estimator outputs make a range bound too conservative.

8.6 Planned estimator workflow

The workflow is shown in Figure 10. It deliberately keeps estimation and validation separate: tail and dependence diagnostics determine candidate estimators; out-of-sample VaR/ES backtests determine whether the added dependence complexity improves risk forecasts.

9 Simulation Design and Projected Diagnostics

This section specifies the planned simulation suite. Its purpose is to verify which theoretical estimator is appropriate under known data-generating mechanisms and to quantify finite-threshold behavior before the real-data analysis. No simulation has yet been executed; numeric entries in the result tables remain **xx** until the experiment runner writes validated CSVs.

9.1 Experiment families

Family I: independent i.n.i.d. tails. This family reproduces the original baseline: Pareto, Lomax, Burr, and Student summands with unequal constants, unequal scales, signed exposures, inactive components, and finite/infinite mean variants. It is designed to verify the maximum constant, sum constant, BRE bound, stratification, and second-order correction.

Family II: common-shock and latent-shock models. Observed variables are generated by $X_i = b_i Z_0 + E_i$, multi-shock factor matrices $X = BZ + E$, sparse loadings, dense market loadings, and mixed signs. This family is designed to verify that the shock-basis constant is correct and to quantify how badly observed-coordinate independence can misattribute risk.

Family III: block dependence. Coordinates are grouped into blocks with within-block common shocks or copulas and independent blocks. This family is designed to test block-tail estimation and optimal block granularity.

Family IV: copula dependence. Margins are Pareto or Lomax and dependence is generated by Gaussian, Student t , Clayton, Gumbel, Frank, and selected vine copulas. This family is designed to evaluate exact copula-kernel CdMC and to compare it with copula importance sampling and independent misspecification.

Family V: MRV spectral models. Radial variables are Pareto or second-order Pareto; angular distributions are axis-concentrated, ray mixtures, logistic/simplex distributions, and empirical resamples. This family is designed to test the spectral integral, radial CdMC, and spectral control variates.

Family VI: hidden-cone models. Mixtures and copulas are calibrated to be asymptotically independent at first order but to possess pair-cone or block-cone hidden mass. This family is designed to test when the hidden-cone term dominates marginal second order.

9.2 Primary metrics

Each design family is planned to report, on completion:

- true or high-precision reference probability $\mu(x)$;
- relative bias of asymptotic approximations;
- empirical relative error of estimators;
- work-normalized relative error (Definition 8.2);
- coverage of confidence intervals (Hoeffding, Bernstein from Theorem 8.8, self-normalized);
- estimator-output tail diagnostics;
- error decomposition by axis, shock, block, spectral, and hidden-cone components;
- sensitivity to threshold, dimension, tail index, and dependence strength.

9.3 Planned result tables

9.4 Reproducibility contract

Every simulation row will contain the generator name and version, master seed, PCG-style spawned-seed per replicate, threshold, number of outer samples, number of inner samples if any, CPU time, peak memory, git hash, config hash, and software versions. The replacement rule is simple: if a result CSV passes schema validation against `results/SCHEMA.md`, the placeholder figure or table may be replaced by a generated PDF/TeX file with the same label and caption. All simulation runs are executed through the `FactorTail` library (<https://github.com/osolari/FactorTail>); the `config_hash` field records the exact library version and the YAML configuration used.

Remark 9.1 (Seeds, reproducibility, and compute budget). Each design family will be run with a master seed and independent PCG-spawned seeds per replicate, so that any single replicate can be re-executed in isolation. The reference (high-precision) tail probability $\mu(x)$ will be computed

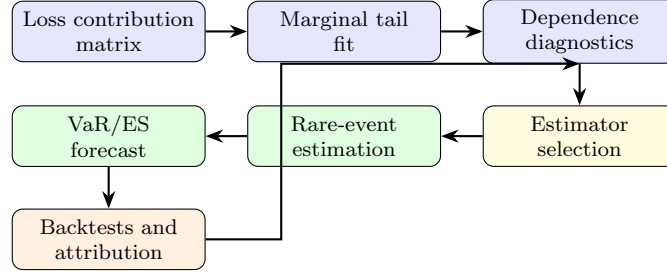


Figure 10: Estimator workflow from raw loss contributions to model diagnostics, rare-event estimation, VaR/ES forecasts, and backtests. The same workflow is used for synthetic experiments and for real data.

Family	Tail model	Dependence model	Dimensions	Thresholds	Planned outputs
I	Pareto, Lomax, Burr, Student	Independent, non-identically distributed	xx	xx	F1, F8, independent simulation table
II	Pareto, Lomax	Common shocks and latent shocks	xx	xx	F11, shock-attribution table, bias table
III	Pareto mixtures	Independent blocks with dependent within-block components	xx	xx	Block-estimator table and cluster diagnostics
IV	Pareto margins	Gaussian, Student t , Archimedean, and vine copulas	xx	xx	Copula-kernel relative error and misspecification bias
V	Pareto radial	Axis, ray, and simplex spectral laws	xx	xx	F12 and spectral BRE metrics
VI	Mixture / hidden RV	Axis first order plus pair or block hidden cones	xx	xx	F13 and hidden-scale comparison

Table 5: **(PARTIAL.)** Simulation design grid. Design columns (experiment, dependence class, estimators tested) are populated; truth-method and outcome columns contain **xx** placeholders until the experiment runner writes the final CSV and table. The design deliberately separates first-order dependence, hidden dependence, second-order marginal behavior, and estimator availability.

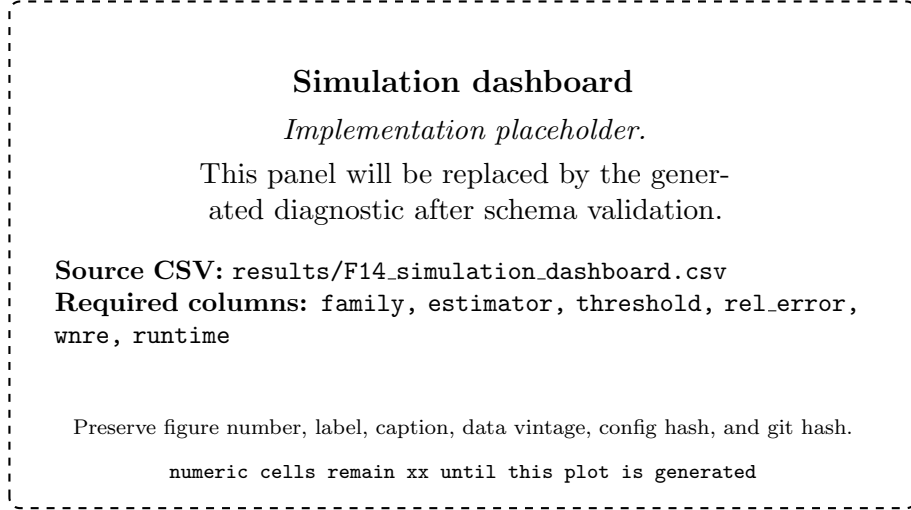


Figure 11: **(PLANNED)**. Simulation dashboard. The generated dashboard will show relative error versus threshold, work-normalized relative error versus dimension, and approximation bias by model family. Source schema: results/F14_simulation_dashboard.csv.

Design	N	α	Threshold	True μ	First-order rel. bias	CdMC rel. error	Runtime
Pareto i.n.i.d.	xx	xx	xx	xx	xx	xx	xx
Lomax finite mean	xx	xx	xx	xx	xx	xx	xx
Burr second order	xx	xx	xx	xx	xx	xx	xx
Signed inac- tive compo- nents	xx	xx	xx	xx	xx	xx	xx

Table 6: **(PLANNED)**. Independent simulation results. Values remain xx until generated. The table will report first-order and second-order accuracy, relative error, and runtime for the independent baseline.

Design	True dependence	Best estimator	Indep. bias	Best RE / WNRE	Attribution
Common shock	xx	xx	xx	xx	xx
Sparse latent shocks	xx	xx	xx	xx	xx
Block copula	xx	xx	xx	xx	xx
Student t copula	xx	xx	xx	xx	xx
MRV ray mixture	xx	xx	xx	xx	xx
Hidden pair cone	xx	xx	xx	xx	xx

Table 7: **(PLANNED)**. Dependent simulation results. Values remain xx until generated. The table will compare independent, copula-kernel, latent-shock, block, spectral, and hidden-cone estimators under known dependence.

either in closed form (Families I, II) or by a high-budget reference estimator (Families III–VI); reference-estimator budgets will be at least an order of magnitude larger than the production estimator budgets. Total compute is bounded by the resources reported in the experiment manifest (Appendix G).

10 Real Data Analysis: Protocol and Projected Outputs

This section is written as a full empirical paper section with placeholders. It is intentionally detailed so that the analysis can be executed without changing the manuscript structure. No empirical run has been executed yet; numerical values remain `xx` until the data pipeline is run. The public-data component uses the Kenneth French Data Library, and the licensed security-level extension uses CRSP through WRDS when institutional access is available [French, 2026, Center for Research in Security Prices, 2026].

10.1 Empirical questions and reporting contract

The empirical study is organized around six questions.

- Q1. Do signed one-sided factor and residual tails exhibit stable regularly varying behavior over rolling windows?
- Q2. Are active loss contributors compatible with a common tail index, or is a dominant-tail or heterogeneous-index approximation required?
- Q3. Do factor and residual extremes look independent, latent-shock driven, block dependent, copula dependent, or MRV/spectral?
- Q4. Which rare-event estimator gives the smallest work-normalized error for one-day VaR and ES under the same fitted tail model?
- Q5. Do heavy-tail forecasts improve out-of-sample VaR/ES calibration relative to Gaussian, historical, and GARCH/EVT baselines?
- Q6. How sensitive are conclusions to thresholds, rolling-window length, portfolio universe, crisis subsamples, and dependence specification?

All empirical claims in the final paper must be traceable to a generated file in `results/`. A table value may remain `xx` in a development draft, but any non-placeholder value must record the data vintage, configuration hash, script hash, random seed when applicable, and validation status. Data specifications, vintage rules, and freezing protocol are in Appendix F. The reference implementation is the `FactorTail` library (<https://github.com/osolari/FactorTail>); the `script_hash` field records the library commit used for each run.

10.2 Data panels, vintages, and freezing rules

The core public data source is the Kenneth French Data Library. The planned panels are daily FF3 factors, daily FF5 factors, daily momentum, daily industry portfolios, size/book-to-market portfolios, operating-profitability and investment portfolios, and the risk-free rate. The licensed extension uses CRSP daily security returns through WRDS, with delisting-return treatment,

Experiment	Required script / output	Figures	Tables	Status
Independent replication	<code>run_independent</code> baseline outputs	F1, F8	Independent simulation table	planned
Common-shock simulation	<code>run_common_shock</code> outputs	F11	Dependent simulation table	planned
Copula-kernel test	<code>run_copula_cdmc</code> outputs	F14	Dependent simulation rows	planned
MRV spectral test	<code>run_spectral_mrv</code> outputs	F12, F14	Spectral rows	planned
Hidden-cone test	<code>run_hidden_rv</code> outputs	F13	Hidden-cone rows	planned
FF public data	<code>run_ff_empirical</code> outputs	F15–F18	Empirical tables	planned
CRSP extension	<code>run_crsp_empirical</code> outputs	F15–F16	Empirical tables	optional / planned

Table 8: Experiment status. This is a project-management artifact inside the paper: it marks each planned experiment, required data product, expected figure/table outputs, and current completion status. The column entries *planned* and *optional/planned* are real categorical labels, not placeholders.

Universe	Model and tail fit	Dependence diagnostics	Estimator candidates	Backtest / sta- tus
Market portfolio	FF5 regression; Hill, GPD	χ , $\bar{\chi}$, spectral mass	Independent, latent, MRV	Window xx; sta- tus xx
25 size–BM portfolios	FF5 panel regressions; Hill/GPD/bootstrap	Pair heatmap, proposed blocks	Independent, block, copula	Window xx; sta- tus xx
30 industry portfolios	FF5 plus industry residual; Hill/GPD	Block spectra, hidden cones	Block, MRV, HRV diagnostic	Window xx; sta- tus xx
CRSP universe	Rolling constituent betas; Hill/GPD/residual LLN	Residual clusters and sector blocks	Latent, block, independent benchmark	Window xx; sta- tus xx
Stress portfolios	High-loading sorts; Hill/GPD	Active-pair tails and crisis windows	Copula, MRV, HRV diagnostic	Window xx; sta- tus xx

Placeholder. Fill from `results/T-empirical_design_matrix.csv`.

Table 9: **(PARTIAL.)** Planned empirical design matrix. Universe, factor model, tail estimator, and estimator-candidate columns are populated; window and run-status columns remain xx until populated by the experiment registry. See Appendix F for vintage controls.

share-code filters, exchange-code filters, and market-equity screens documented in the data appendix (Appendix F).

Freezing rules. Each empirical run records the data vintage (download timestamp and source file hash), the analysis configuration (config hash and git commit), and the random seed. Once a run is included in the manuscript, its inputs are frozen: re-running the analysis must reproduce the same outputs bit-for-bit (subject to documented platform-tolerance bounds). Re-fits with updated data are reported as separate runs and never overwrite existing rows.

10.3 Portfolio construction and factor fits

The analysis uses four portfolio families:

1. canonical test portfolios from the French library;
2. equal-weighted and value-weighted industry portfolios;
3. synthetic long-only and long-short portfolios formed by random or optimized exposure vectors;
4. CRSP security-level portfolios when licensed access is available.

For each portfolio p , rolling windows of length $w \in \{\mathbf{xx}, \mathbf{xx}, \mathbf{xx}\}$ fit FF3, FF5, and FF5+momentum models:

$$r_{p,t} = \alpha_{p,t} + \beta_{p,t}^\top f_t + \varepsilon_{p,t}.$$

The loss contribution vector is

$$X_{p,t} = (-\beta_{1,t}f_{1,t}, \dots, -\beta_{d,t}f_{d,t}, -\varepsilon_{p,t}).$$

This vector is the input to marginal, dependent-kernel, latent-shock, block, and spectral estimators.

10.4 Marginal tail analysis

For every signed factor and residual contribution, the pipeline estimates one-sided right-tail indices using Hill, Pickands, and POT/GPD estimators across threshold grids. It will report stability plots, common-index tests, threshold selection, and uncertainty bands. No winsorization is applied to the tail sample used for risk estimation; data-cleaning flags are reported separately.

10.5 Dependence diagnostics

Dependence is reported at three levels. First, ordinary dependence: Pearson, Spearman, Kendall, and rolling correlations of signed loss contributions. Second, extremal dependence: empirical $\chi(u)$, $\bar{\chi}(u)$, Ledford–Tawn η , pair exceedance rates, and co-exceedance stability. Third, geometric dependence: empirical spectral mass on axes, rays, sectors, and hidden cones.

Panel	Source	Date range	Assets / series	Vintage and audit fields
FF3 daily factors	Kenneth French DL	xx-xx	xx	checksum xx; access xx
FF5 daily factors	Kenneth French DL	xx-xx	xx	checksum xx; access xx
Momentum daily	Kenneth French DL	xx-xx	xx	checksum xx; access xx
Industry portfolios	Kenneth French DL	xx-xx	xx	checksum xx; access xx
Characteristic portfolios	Kenneth French DL	xx-xx	xx	checksum xx; access xx
CRSP securities	WRDS/CRSP licensed	xx-xx	xx	filters xx; access xx

Table 10: **(PLANNED)**. Real-data panels and data-vintage controls. All numerical entries are xx until the data pull is executed. The final version will report date range, number of assets or portfolios, missingness, source vintage, and access timestamp.

Contribution	Sign	Threshold	Hill $\hat{\alpha}$	POT $\hat{\alpha}$	CI	Common-index decision
Market	loss side	xx	xx	xx	xx	xx
SMB	loss side	xx	xx	xx	xx	xx
HML	loss side	xx	xx	xx	xx	xx
RMW	loss side	xx	xx	xx	xx	xx
CMA	loss side	xx	xx	xx	xx	xx
Momentum	loss side	xx	xx	xx	xx	xx
Residual	loss side	xx	xx	xx	xx	xx

Table 11: **(PLANNED)**. Tail-index estimates for signed factor and residual contributions. Values are xx placeholders. The final table will report Hill, Pickands, and POT estimates, selected thresholds, standard errors or bootstrap intervals, and common-index diagnostic outcomes.

Pair/block	Kendall	χ	$\bar{\chi}$	η	Spectral class	Decision
Market-SMB	xx	xx	xx	xx	xx	xx
Market-HML	xx	xx	xx	xx	xx	xx
Market-Residual	xx	xx	xx	xx	xx	xx
Value block	xx	xx	xx	xx	xx	xx
Profitability/investment block	xx	xx	xx	xx	xx	xx
Industry cluster 1	xx	xx	xx	xx	xx	xx

Table 12: **(PLANNED)**. Tail-dependence diagnostics for factor and residual loss contributions. Values are xx placeholders. The final table will identify whether each pair is consistent with asymptotic dependence, tail dependence, common-shock behavior, or hidden-cone dependence.

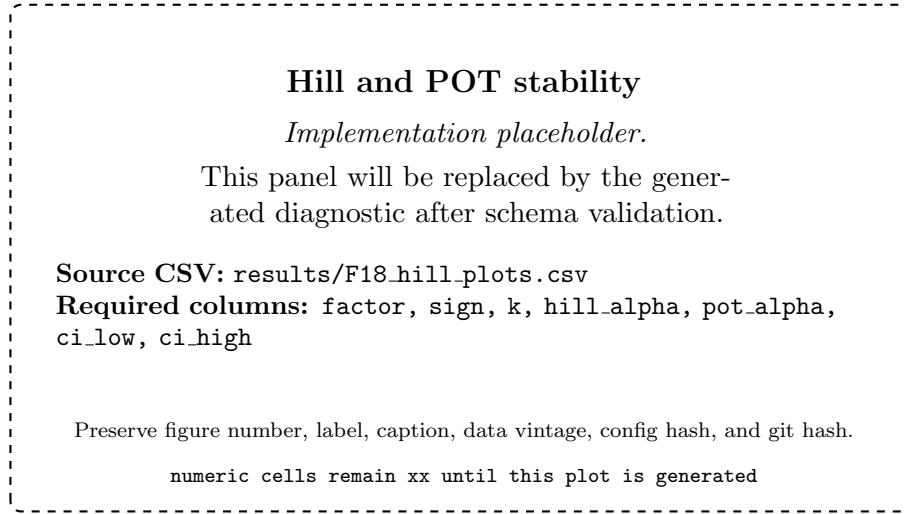


Figure 12: **(PLANNED)**. Hill and POT stability. The generated figure will show tail-index estimates across order-statistic thresholds for signed factors and residuals, with selected thresholds and bootstrap bands. Source schema: results/F18_hill_plots.csv.

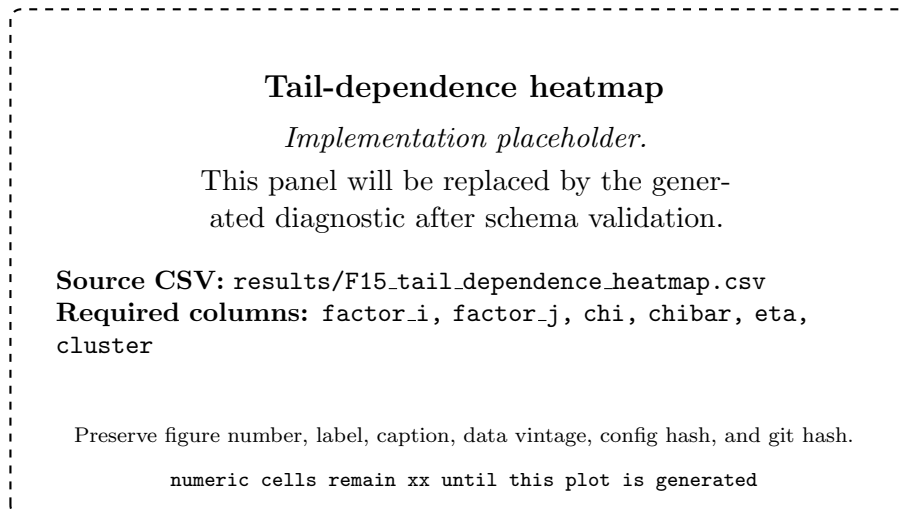


Figure 13: **(PLANNED)**. Tail-dependence heatmap. The generated figure will show pairwise χ , $\bar{\chi}$, or η estimates for signed factor and residual contributions, with clustering used to propose block structures. Source schema: results/F15_tail_dependence_heatmap.csv.

10.6 Rare-event estimation on real data

For each portfolio, date, model, and confidence level, the pipeline estimates

$$\mu_t(x) = \mathbb{P}_t(L_{p,t+1} > x)$$

using the following estimators: independent CdMC, dependent copula-kernel CdMC, latent-shock CdMC from PCA/ICA or fitted common-shock models, block CdMC from clustered tail-dependence blocks, spectral CdMC from empirical angular resampling, and filtered historical simulation. Thresholds are selected to correspond to VaR levels 99%, 99.5%, and 99.9%, plus a stress grid beyond observed quantiles for estimator-efficiency comparison.

10.7 VaR and ES forecasting

The VaR forecast $\hat{q}_{p,t}(\tau)$ solves

$$\hat{\mathbb{P}}_t(L_{p,t+1} > \hat{q}_{p,t}(\tau)) = 1 - \tau.$$

The ES forecast is computed either by tail integration of the fitted rare-event curve or by conditional-excess simulation under the same estimator. The final paper will report forecast paths, exception indicators, ES exceedance residuals, and crisis-period decompositions.

10.8 Backtesting

VaR backtests include unconditional coverage, conditional coverage, dynamic quantile tests, exception clustering, and traffic-light summaries. ES backtests include Acerbi–Szekely-style exceedance residual tests and comparative loss functions. The final analysis will report both full-sample and rolling-window statistics, with crisis-window attribution separated from ordinary periods.

10.9 Crisis-window attribution

For each stress window, the paper will report the estimated contribution of axes, latent shocks, blocks, spectral sectors, and hidden cones to the tail forecast. Planned windows include the 1987 crash if the selected data panel begins early enough, the dot-com unwind, the global financial crisis, the COVID-19 shock, and the 2022 inflation/rate-hike period. The exact windows are fixed in the configuration file and reported in the data appendix.

10.10 Planned real-data experiment registry

The execution plan separates model validation from estimator-efficiency measurement. Experiments that compare estimators hold the fitted distributional model fixed. Experiments that compare fitted models use the same forecasting calendar, loss definition, threshold grid, and backtest protocol.

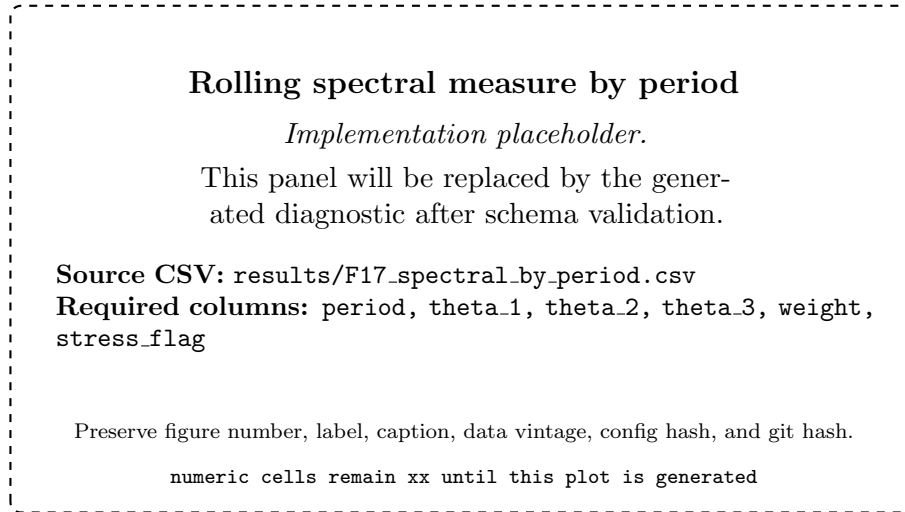


Figure 14: **(PLANNED)**. Rolling empirical spectral measure. The generated figure will compare angular mass before, during, and after stress windows, showing whether extremes are axis-dominated, common-shock dominated, or spread over a sector of the simplex. Source schema: results/F17_spectral_by_period.csv.

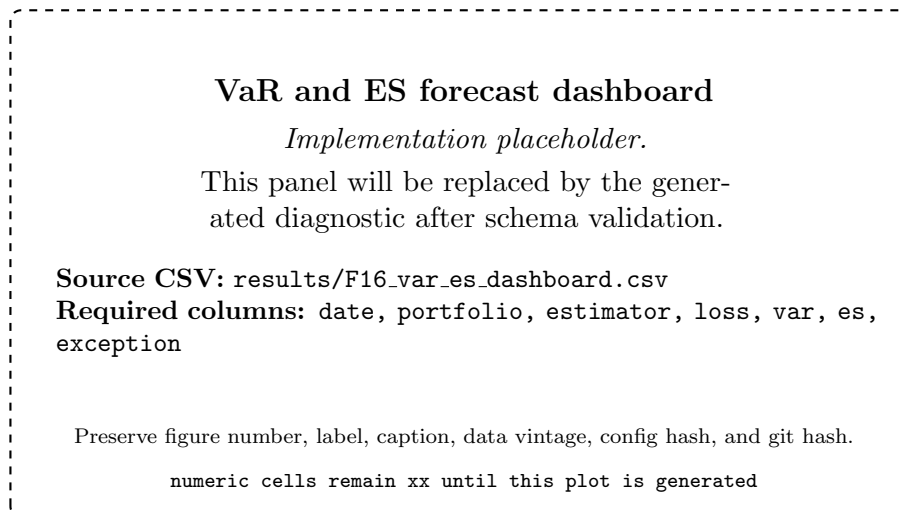


Figure 15: **(PLANNED)**. VaR/ES forecast dashboard. The generated figure will overlay realized portfolio losses, rolling VaR forecasts, ES forecasts, and exception markers for each estimator. Panels will include full sample and stress-window zooms. Source schema: results/F16_var_es_dashboard.csv.

Estimator	VaR level	Exceptions	Kupiec p	CC p	ES test p	Score rank
Independent CdMC	xx	xx	xx	xx	xx	xx
Latent-shock CdMC	xx	xx	xx	xx	xx	xx
Block CdMC	xx	xx	xx	xx	xx	xx
Copula-kernel CdMC	xx	xx	xx	xx	xx	xx
Spectral CdMC	xx	xx	xx	xx	xx	xx
Historical simulation	xx	xx	xx	xx	xx	xx
Parametric t	xx	xx	xx	xx	xx	xx

Table 13: **(PLANNED)**. VaR and ES backtest results. Values are xx placeholders. The final table will compare independent, latent-shock, block, copula, spectral, historical, and parametric benchmarks across exception rate, independence, conditional coverage, ES residual, and comparative scoring metrics.

Window	Axis share	Latent shock share	Block share	Spectral sector share	Hidden-cone share	Dominant driver
1987 crash window	xx	xx	xx	xx	xx	xx
Dot-com un-wind	xx	xx	xx	xx	xx	xx
Global financial crisis	xx	xx	xx	xx	xx	xx
COVID-19 shock	xx	xx	xx	xx	xx	xx
2022 rate/inflation period	xx	xx	xx	xx	xx	xx

Table 14: **(PLANNED)**. Crisis-window tail attribution. Values are xx placeholders. The final table will decompose forecast tail probability into marginal axes, latent common shocks, blocks, spectral sectors, and hidden cones by stress period.

Experiment	Target	Estimators	Forecast levels	Output files	Status
E1 tail diagnostics	signed factor/residual tails	Hill, Pickands, POT	n/a	F11, T_tail_indices	xx
E2 dependence diagnostics	factor/residual joint tails	chi, chibar, eta, spectral mass	n/a	F12, F13, T_dependence_diagnostics	xx
E3 independent baseline	one-day portfolio loss	CrMC, CdMC, stratified, CV	0.99, 0.995, 0.999	T_var_es_comparison, F9	xx
E4 latent-shock model	common-shock loss tail	latent CdMC, common-shock formula	0.99, 0.995, 0.999	F14, T_model_gate	xx
E5 copula/block robustness	dependent factor loss tail	copula CdMC, block CdMC, radial CdMC	0.99, 0.995, 0.999	F15, T_compute_cost	xx
E6 backtesting	realized losses vs forecasts	all retained models + baselines	0.99, 0.995	T_backtest, F10	xx
E7 crisis windows	stress-event stability	retained production models	0.99, 0.995, 0.999	T_crisis_windows	xx

Table 15: **(PARTIAL.)** Planned real-data experiment registry. Experiment descriptors, estimator candidates, forecast levels, and output identifiers are populated; the status/outcome column remains xx until the run registry is populated.

10.11 Runtime and implementation cost

The empirical section will report wall-clock time, number of random variates, number of conditional-kernel evaluations, fit time, and memory use. Runtime will be reported separately for training, diagnostic estimation, and daily forecasting.

10.12 Empirical workflow diagram

11 Discussion and Development Roadmap

11.1 Theoretical results established versus empirical claims projected

The expanded manuscript separates two kinds of statements.

Theoretical (established under the stated assumptions). The independent finite- N catastrophe principle (Theorem 4.2), the single-big-jump sum equivalence (Theorem 4.4), the corrected second-order expansion with leave-one-out mean shift (Theorem 4.6), independent-CdMC unbiasedness and the scale-invariant BRE bound $N^\alpha - 1$ (Proposition 8.3), the arbitrary-dependence CdMC identity (Theorem 5.1), the conditional-tail-envelope BRE result (Proposition 8.4), the latent-shock tail constant and BRE bound (Theorem 5.6 and Proposition 8.5), the independent-block reduction (Theorem 5.8), the MRV linear-risk asymptotic (Theorem 6.4) with the spectral CdMC identity and BRE (Propositions 6.5 and 8.6), the hidden-cone second-order expansion (Theorem 7.3), the exact and asymptotic control-variate VRE result (Proposition 8.7), and the Bernstein-type confidence interval (Theorem 8.8). Each of these is proved in Appendices B to D with the underlying assumptions explicit.

Projected (planned, not yet executed). All simulation diagnostics in Section 9, all real-data diagnostics, forecasts, and backtests in Section 10, and every figure with a (PLANNED) caption prefix or table cell marked **xx**. Numerical performance, backtest success, and production runtime conclusions are not claimed until the corresponding pipeline outputs replace those entries.

11.2 Limitations and counterexamples

The BRE bound $N^\alpha - 1$ is finite for any finite N but diverges as $N \rightarrow \infty$. The independent CdMC is therefore not suited as-is to high-dimensional limit regimes; in such regimes the latent-shock CdMC (Proposition 8.5) replaces N by the number of active shocks K , and the spectral CdMC (Proposition 8.6) replaces N by a spectral integral that need not grow with the ambient dimension.

The dependent CdMC identity (Theorem 5.1) is exact only when the conditional kernels $p_i(t; X_{-i})$ are evaluated exactly. A misspecified copula or a poorly fit conditional kernel produces a biased estimator at every threshold, and the bias can be much larger than the Monte Carlo standard error. Remarks 5.5 and 5.9 discuss identifiability and stability mitigations; the empirical protocol of Section 10 reports model-residual diagnostics before any kernel estimator is trusted.

Figure	Primary diagnostic	Source and replacement rule
F11	rolling left/right tail indices	Tail-index stability CSV; replace after threshold-selection validation.
F12	empirical angular/spectral mass	Spectral-mass CSV; replace after MRV diagnostic run.
F13	pairwise tail-dependence heat map	Tail-dependence CSV; replace with heat map and confidence overlay.
F14	latent-shock loading stability	Shock-loading CSV; replace with rolling eigen/ICA loading plot.
F15	model-gate timeline	Model-selection CSV; replace with estimator-gate timeline.
F16	VaR/ES forecast comparison	Forecast CSV; replace with multi-model VaR/ES forecast path.
F17	violation clusters	Backtest-hit CSV; replace with hit process and rolling violation rate.
F18	runtime frontier	Runtime CSV; replace with runtime-versus-error frontier plot.

Table 16: Planned real-data figure replacement registry. Each placeholder figure has a source CSV and replacement rule so that visual results can be regenerated without changing labels or captions. The cells in this table are real categorical descriptors.

Estimator	Fit time	Forecast time/date	Replicate cost	WNRE	Notes
Independent CdMC	xx	xx	xx	xx	xx
Latent-shock CdMC	xx	xx	xx	xx	xx
Block CdMC	xx	xx	xx	xx	xx
Copula-kernel CdMC	xx	xx	xx	xx	xx
Spectral CdMC	xx	xx	xx	xx	xx
Hidden-cone mixture	xx	xx	xx	xx	xx

Table 17: **(PLANNED)**. Runtime and computational-cost placeholder. Values are xx placeholders. The final table will report per-date forecast cost, per-replicate cost, total runtime, and work-normalized relative error for each estimator.

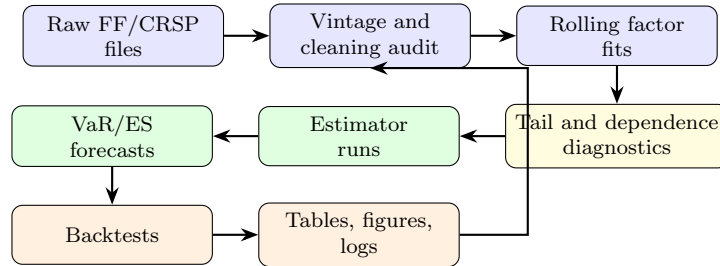


Figure 16: Real-data workflow from raw factor and return files through tail diagnostics, dependence diagnostics, estimator selection, VaR/ES forecasting, backtesting, and artifact generation. Every output file is validated against the schema in `results/SCHEMA.md`.

Hidden regular variation (Definition 7.1, Assumption 7.2, and Theorem 7.3) requires identification of the hidden tail scale $\overline{H}_2$ from finite samples. In small samples, threshold noise can be mistaken for a hidden scale. The manuscript treats hidden-cone results as a second-order layer and requires threshold-stability diagnostics (Appendix D) before any hidden-cone estimator enters a primary VaR/ES forecast.

11.3 Connection to the literature placement

The independent finite- N sharpening of Section 4 positions this paper relative to Pourbabaee and Solari [2019] in the form spelled out in Section 2.7: scale-invariant BRE bound, corrected second-order expansion with $N = 1$ specialization, and explicit assumption stack. The dependent extensions of Sections 5 to 7 connect to Asmussen and Kroese [2006], Hartinger and Kortschak [2009], Blanchet and Rojas-Nandayapa [2011], Cheng et al. [2025] (rare-event simulation under dependence), Hult et al. [2005], Resnick [2007], Samorodnitsky and Sun [2016] (MRV foundations), and Ledford and Tawn [1996], Maulik and Resnick [2005], Das et al. [2013] (hidden regular variation). The empirical protocol of Section 10 follows the factor-portfolio backtesting tradition of McNeil et al. [2015], Glasserman et al. [2002] and adds dependence-diagnostic gating for estimator selection.

11.4 Recommended development sequence

The recommended implementation sequence is:

1. finalize and test the independent baseline replication;
2. implement common-shock and latent-shock simulation;
3. implement empirical marginal tail diagnostics;
4. implement pairwise tail-dependence and spectral diagnostics;
5. implement latent, block, and spectral estimators;
6. run real-data rolling VaR/ES forecasts;
7. add hidden-cone estimators only after diagnostics justify them.

11.5 Conclusion

The original independent-factor paper is strongest when the extreme event is well represented by one large observed coordinate. The financial problem asks for more: extremes may be generated by one latent market shock, one dependent block, one radial vector movement, or one hidden joint-tail cone. The expanded paper keeps the original single-big-jump insight but moves it to the correct object. In the independent model that object is a coordinate; in the latent model it is a shock; in MRV it is a spectral/radial movement; and in hidden regular variation it is a slower-scale joint cone. The accompanying empirical protocol is designed to determine which object is visible in real financial data and whether modeling it improves VaR, ES, attribution, and simulation efficiency. The full method is implemented as the open-source **FactorTail** library

(<https://github.com/osolari/FactorTail>); the repository is the canonical reference for any future numerical or empirical claim downstream of this manuscript.

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A Notation and Output Conventions

This appendix consolidates the notation used throughout the paper. Symbols are grouped by section of first use. When a symbol is reused with a different meaning in a later section the alternative usage is noted explicitly.

A.1 Probability and indicator conventions

$\mathbb{P}$ and $\mathbb{E}$ denote the probability and expectation under a generic probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carrying all random objects of interest. Indicator functions are written $\mathbf{1}\{A\}$ or $\mathbf{1}_A$. For a univariate distribution function F we write $\overline{F} = 1 - F$ for the survival function. For a multivariate cumulative distribution function or measure H we write $\overline{H}$ for the tail object specified locally. Independence of random vectors is written $\perp\!\!\!\perp$. The symbol $\sim$ denotes asymptotic equivalence: $f(x) \sim g(x)$ means $f(x)/g(x) \rightarrow 1$ as $x \rightarrow \infty$. The symbol $\Rightarrow$ denotes weak convergence of measures or distributions.

A.2 Loss vector, sums, and selected maxima

Symbol	Meaning
$X = (X_1, \dots, X_N)$	Observed N -vector of loss contributions; positive values are losses.
S_N	Aggregate loss $\sum_{i=1}^N X_i$.
M_N	Maximum contribution $\max_{i=1}^N X_i$.
S_{-i}	Leave-one-out sum $\sum_{j \neq i} X_j$; $S_{-i} = 0$ if $N = 1$.
M_{-i}	Leave-one-out maximum $\max_{j \neq i} X_j$; $M_{-i} = -\infty$ if $N = 1$.
$T_i(x)$	Selected-maximum conditioning threshold $(x - S_{-i}) \vee M_{-i}$.
$R_i(X) = 1$	Event that coordinate i is the selected maximum under a fixed deterministic tie rule.
$\mu(x)$	Rare-event probability $\mathbb{P}(S_N > x)$.
μ_{-i}	Leave-one-out mean $\sum_{j \neq i} \mathbb{E}X_j$.

A.3 Regular variation and active-set constants

Symbol	Meaning
$\text{RV}_{-\alpha}$	Class of regularly varying tail functions with index $-\alpha$.
$\overline{G}$	Common reference survival function with $\overline{G} \in \text{RV}_{-\alpha}$.
α	Tail index, $\alpha > 0$; when context permits, $\alpha > 1$ for first-moment finiteness.
c_i	Positive-tail equivalence constant: $\mathbb{P}(X_i > x) \sim c_i \overline{G}(x)$.
$\mathcal{A}$	Active right-tail set $\{i : c_i > 0\}$.
C	First-order independent constant $\sum_{i \in \mathcal{A}} c_i$.
c_i^+, c_i^-	Signed-tail constants for the positive and negative tails of X_i .
p_i^+, p_i^-	Signed-factor-tail constants used in the financial loss map.
b_i	Loss-direction coefficient with $X_i = b_i F_i$.
$a \in \mathbb{R}^d$	Portfolio loading vector; $L = a^\top F$ is the loss after sign alignment.
$A(x)$	Second-order auxiliary function in the marginal expansion.
ρ	Second-order regular-variation parameter $\rho \leq 0$.
d_i	Marginal second-order coefficient appearing in Assumption 3.5.
$D_{\text{marg}}, D_{\text{mean}}$	Marginal and mean-shift second-order constants.
D_{hid}	Hidden-cone second-order constant.
$r(x)$	Higher-order remainder scale satisfying $r(x) = o(\overline{G}(x)A(x) + \overline{G}(x)/x)$.

A.4 Estimators and efficiency notions

Symbol	Meaning
CrMC	Crude Monte Carlo.
CdMC	Conditional Monte Carlo (Asmussen–Kroese style).
BRE	Bounded relative error.
VRE	Vanishing relative error.
Z^{CrMC}	Crude indicator estimator $\mathbf{1}\{S_N > x\}$.
Z^{CdMC}	Summed conditional Monte Carlo estimator $\sum_i p_i(T_i; X_{-i})$.
Z^{strat}	Stratified conditional Monte Carlo estimator.
Z^{ana}	Closed-form analytic-marginal estimator.
Z^{VRE}	Control-variate vanishing-relative-error estimator.
Z^{spec}	Spectral/radial conditional Monte Carlo estimator.
$W(\hat{\mu})$	Work-normalized variance $\text{Var}(\hat{\mu}) \cdot \text{cost}(\hat{\mu})$.
$\text{eff}(\hat{\mu}_1, \hat{\mu}_2)$	Efficiency ratio $W(\hat{\mu}_2)/W(\hat{\mu}_1)$.
$\hat{\mu}_n, \text{CI}_n$	Sample mean estimator and confidence interval from n replicates.

A.5 Dependence classes and conditional kernels

Symbol	Meaning
$p_i(t; X_{-i})$	Conditional tail kernel $\mathbb{P}(X_i > t \mid X_{-i})$.
$F = BZ + \varepsilon$	Latent-shock factor representation; B is the loading matrix, Z the shock vector.
$B \in \mathbb{R}^{d \times K}$	Latent-shock loading matrix.
$q = B^\top a$	Portfolio exposure to latent shocks.
Z_k	Latent shock k ; independent across k in the latent-shock model.
Π	Block partition of the index set $\{1, \dots, N\}$.
$\mathcal{B}_b$	Block b of the partition Π .
$C(u_1, \dots, u_d)$	Copula coupling the marginals of F .

A.6 Multivariate and hidden regular variation

Symbol	Meaning
MRV	Class of multivariate regularly varying distributions.
HRV	Class of hidden-regular-variation distributions.
ν	MRV tail (limit) measure on the cone $\mathcal{E}_0 = [0, \infty)^d \setminus \{0\}$.
ν_2	Hidden-cone tail measure on $\mathcal{E}_2 \subsetneq \mathcal{E}_0$.
$\mathcal{E}_0, \mathcal{E}_2$	Outer and hidden cones used in the MRV/HRV decomposition.
$S(\cdot)$	Spectral (angular) measure on the unit sphere or simplex.
Θ	Angular random vector with law S .
R	Radial component with regularly varying survival.
$\overline{H}_2(x)$	Hidden tail scale satisfying $\overline{H}_2(x) = o(\overline{G}(x))$.

Symbol	Meaning
$\chi(q), \bar{\chi}(q), \eta$	Coefficients of tail dependence, complementary tail dependence, and Ledford–Tawn residual index.

A.7 Financial and empirical objects

Symbol	Meaning
$F \in \mathbb{R}^d$	Factor return vector.
$L = a^\top F$	Portfolio loss after sign alignment.
$\text{VaR}_\tau(L)$	Value-at-risk of L at confidence level τ .
$\text{ES}_\tau(L)$	Expected shortfall of L at level τ .
w	Rolling estimation window length.
τ	VaR/ES confidence level (e.g., $\tau \in \{0.99, 0.995, 0.999\}$).

A.8 Output conventions and placeholder rule

Symbol	Meaning
xx	Placeholder numeric value; not an empirical result.

Placeholder convention. Every table cell containing **xx** is deliberately unfilled. It must be replaced only by a value generated from a validated CSV with a recorded configuration hash, data vintage, and run identifier (see Appendix G). Placeholder figures are TikZ instruction sheets whose captions describe the intended diagnostic and source schema. A placeholder is not evidence for or against any model.

B Proofs for the Independent Baseline

This appendix collects the full proofs of the results stated in Section 4. Each proof environment is labelled by the result it establishes.

Proof of Theorem 4.2. Let $A_i(x) = \{X_i > x\}$. By finite inclusion–exclusion,

$$\sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) \leq \mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

Under Assumption 4.1 the variables are independent, so

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(X_i > x)\mathbb{P}(X_j > x) \sim c_i c_j \bar{G}(x)^2 = o(\bar{G}(x)).$$

There are finitely many pairs, so $\sum_{i < j} \mathbb{P}(A_i \cap A_j) = o(\overline{G}(x))$. Higher-order intersections are bounded above by pair intersections and so are also $o(\overline{G}(x))$. Combined with the marginal sum $\sum_i \mathbb{P}(X_i > x) \sim C\overline{G}(x)$, this yields $\mathbb{P}(M_N > x) \sim C\overline{G}(x)$. $\square$

Proof of Theorem 4.4. Let $h = h(x) \uparrow \infty$ with $h = o(x)$ as in Assumption 4.3. Upper bound. Decompose

$$\{S_N > x\} \subseteq \bigcup_{i=1}^N \{X_i > x - h\} \cup \left\{ S_N > x, \max_i X_i \leq x - h \right\}.$$

The first term satisfies

$$\mathbb{P}\left(\bigcup_i \{X_i > x - h\}\right) \sim \sum_i \mathbb{P}(X_i > x - h) \sim \sum_i \mathbb{P}(X_i > x),$$

by inclusion–exclusion and the local invariance $\mathbb{P}(X_i > x - h) \sim \mathbb{P}(X_i > x)$ of Assumption 4.3. The second term is excluded by the truncation conditions in Assumption 4.3: the joint event that two or more truncated components exceed order h is $o(\overline{G}(x))$, and the residual non-large contribution is $o(\overline{G}(x))$ by the same truncation.

Lower bound. With the same h ,

$$\mathbb{P}(S_N > x) \geq \sum_i \mathbb{P}(X_i > x + h, S_{-i} > -h) - o(\overline{G}(x)),$$

where the overlap correction is $o(\overline{G}(x))$ by the pair-exceedance estimate of Theorem 4.2. Independence gives

$$\mathbb{P}(X_i > x + h, S_{-i} > -h) = \mathbb{P}(X_i > x + h) \mathbb{P}(S_{-i} > -h) \sim \mathbb{P}(X_i > x),$$

since $\mathbb{P}(S_{-i} \leq -h) = o(1)$ and $\mathbb{P}(X_i > x + h) \sim \mathbb{P}(X_i > x)$. Summing yields the matching lower bound. $\square$

Proof of Theorem 4.6. Set $r(x) = \overline{G}(x)(|A(x)| \vee x^{-1})$. The events $\{i \text{ selected}\} = \{R_i(X) = 1\}$ partition $\{S_N > x\}$ under the fixed tie-breaking rule of Assumption 3.3. On the i th stratum

$$\mathbb{P}(S_N > x, R_i(X) = 1) = \mathbb{E}[\mathbb{P}(X_i > T_i(x) \mid X_{-i}) \mathbf{1}\{T_i(x) > M_{-i}\}].$$

By Assumption 4.3 the contribution to the right-hand side from the event $\{T_i(x) \leq M_{-i}\}$ is $o(r(x))$, and the same truncation gives negligible mass on $\{|S_{-i}| > h(x)\}$. It therefore suffices to expand $\mathbb{E}[\mathbb{P}(X_i > x - S_{-i})]$ on $\{|S_{-i}| \leq h(x)\}$ and then take expectations.

By Assumption 3.5, locally uniformly on $|S_{-i}| \leq h(x)$,

$$\mathbb{P}(X_i > x - S_{-i}) = \overline{G}(x) \left[c_i + c_i d_i A(x) + \frac{\alpha c_i S_{-i}}{x} + o(|A(x)| + x^{-1}) \right].$$

Taking expectations,

$$\mathbb{E}\mathbb{P}(X_i > x - S_{-i}) = c_i \overline{G}(x) + c_i d_i \overline{G}(x) A(x) + \frac{\alpha c_i \overline{G}(x)}{x} \mathbb{E} S_{-i} + o(r(x)),$$

and $\mathbb{E}S_{-i} = \mu_{-i} = \sum_{j \neq i} \mathbb{E}X_j$ by Assumption 3.5. Summing over active i and discarding inactive contributions (which are lower order by Assumption 3.5) yields the expansion in Theorem 4.6.

Consistency check at $N = 1$. If $N = 1$, then $S_{-1} = 0$ and $\mu_{-1} = 0$; the mean-shift term vanishes, the marginal second-order term reduces to $c_1 d_1 \bar{G}(x)A(x)$, and the formula collapses to the standard single-variable second-order expansion. $\square$

Independent CdMC: unbiasedness and BRE bound. Unbiasedness. For continuous variables the smallest-index tie-breaking rule of Assumption 3.3 gives the disjoint union

$$\{S_N > x\} = \bigsqcup_{i=1}^N \{S_N > x, R_i(X) = 1\}.$$

Conditional on X_{-i} , the i th stratum equals $\{X_i > T_i(x)\} \cap \{T_i(x) > M_{-i}\}$, and independence yields $\mathbb{P}(X_i > T_i(x) \mid X_{-i}) = \bar{F}_i(T_i(x))$. Taking expectations and summing over i gives $\mathbb{E}Z^{\text{ind}}(x) = \mathbb{P}(S_N > x)$.

Scale-invariant BRE bound. We claim $T_i(x) \geq x/N$ almost surely: suppose $M_{-i} < x/N$; then if also $x - S_{-i} < x/N$, one has $S_{-i} > (N-1)x/N$. Since S_{-i} is a sum of $N-1$ terms each bounded by $M_{-i} < x/N$, this gives $S_{-i} \leq (N-1)x/N$, a contradiction. Hence $T_i(x) \geq x/N$, and therefore

$$0 \leq Z^{\text{ind}}(x) \leq B(x) := \sum_{i=1}^N \bar{F}_i(x/N).$$

Since $\mathbb{E}Z^{\text{ind}}(x) = \mu(x)$,

$$\text{Var } Z^{\text{ind}}(x) \leq \mathbb{E}[Z^{\text{ind}}(x)^2] - \mu(x)^2 \leq B(x)\mu(x) - \mu(x)^2.$$

Regular variation gives $\bar{F}_i(x/N)/\bar{F}_i(x) \rightarrow N^\alpha$ and the sum equivalence (Theorem 4.4) gives $\sum_i \bar{F}_i(x) \sim C\bar{G}(x)$, so $B(x)/\mu(x) \rightarrow N^\alpha$. Consequently

$$\limsup_{x \rightarrow \infty} \frac{\text{Var } Z^{\text{ind}}(x)}{\mu(x)^2} \leq N^\alpha - 1.$$

The bound is scale-invariant: replacing X by λX leaves N , α , and the ratio unchanged. $\square$

B.1 Verification notes

Lomax second-order constant. For the Lomax/Pareto-type benchmark $\bar{F}_i(t) = (1+t/\beta_i)^{-\alpha}$ on $t > 0$, $\mathbb{E}X_i = \beta_i/(\alpha-1)$ when $\alpha > 1$. A direct substitution verifies the cross-term in the second-order expansion of $\mathbb{P}(X_i > x - S_{-i})$: expanding to second order in S_{-i}/x gives

$$(1 + (x - S_{-i})/\beta_i)^{-\alpha} = (x/\beta_i)^{-\alpha} (1 - \beta_i/x)^{-\alpha} (1 + S_{-i}/(x - \beta_i))^\alpha,$$

and expansion to first order in S_{-i}/x and to leading order in the marginal-tail auxiliary $A(x) = \beta_i/x$ recovers $c_i d_i A(x) + \alpha c_i S_{-i}/x$ with the same sign convention used in Theorem 4.6. This is the regression check used by the implementation pipeline.

Scale-invariance check. The BRE bound $N^\alpha - 1$ is invariant under $X_i \rightarrow \lambda X_i$ because both $T_i(x)$ and x scale by λ , leaving $\bar{F}_i(T_i(\lambda x)) = \bar{F}_i(T_i(x))$ when $\bar{F}_i$ is correctly rescaled. The bound therefore depends only on N and α , not on the marginal scale constants c_i .

C Proofs for Dependent Extensions

This appendix proves the results stated in Section 5 and Section 6. Each proof environment is labelled by the result it establishes.

Proof of Theorem 5.1. Let

$$E_i(x) = \{S_N > x, R_i(X) = 1\},$$

where R_i is the selected-maximum rule of Assumption 3.3. The events $E_i(x)$ are disjoint and their union is $\{S_N > x\}$. Conditional on X_{-i} , the event $E_i(x)$ is equivalent to X_i exceeding both the sum threshold and the selected-maximum threshold. In the continuous case this is

$$X_i > (x - S_{-i}) \vee M_{-i} = T_i(x).$$

For discrete laws, replacing $> M_{-i}$ by the tie-rule endpoint gives the same identity with an adjusted threshold. Hence

$$\mathbb{P}(E_i(x)) = \mathbb{E}[\mathbb{P}(X_i > T_i(x) \mid X_{-i})] = \mathbb{E}[p_i(T_i(x); X_{-i})].$$

Summing over i proves the identity and the unbiasedness of $Z^{\text{dep}}(x)$. No independence step is required. $\square$

Remark C.1 (Population identity versus estimator). Theorem 5.1 contains two distinct statements. The population-level identity $\mu(x) = \sum_i \mathbb{E}[p_i(T_i(x); X_{-i})]$ holds without any computability assumption on the kernels. The estimator statement $\mathbb{E}Z^{\text{dep}}(x) = \mu(x)$ holds provided the kernels are evaluated exactly per replicate; if the kernels are evaluated by a noisy or approximate routine, an additional bias term appears that is controlled by the kernel approximation error.

Proof of Proposition 5.4. By Theorem 5.1, $\mathbb{E}Z^{\text{dep}}(x) = \mu(x)$. The selected-maximum threshold satisfies $T_i(x) \geq x/N$ almost surely (the argument is the same as in the independent CdMC proof in Appendix B). By Assumption 5.3, for all $t \geq x/N$,

$$p_i(t; X_{-i}) \leq K_{i,x} b_i(t),$$

and therefore

$$Z^{\text{dep}}(x) \leq \sum_{i=1}^N K_{i,x} b_i(x/N).$$

The second-moment envelope in Assumption 5.3 gives

$$\mathbb{E}[Z^{\text{dep}}(x)^2] = O(\mu(x)^2),$$

hence

$$\text{Var } Z^{\text{dep}}(x) \leq \mathbb{E} Z^{\text{dep}}(x)^2 = O(\mu(x)^2),$$

which is bounded relative error. $\square$

Proof of Theorem 5.6. Set $Y_k = q_k Z_k$. Independence of the shocks transfers to the signed shock contributions Y_k . Their right-tail constants are

$$\mathbb{P}(Y_k > x) = \mathbb{P}(q_k Z_k > x) \sim |q_k|^\alpha \{p_k^+ \mathbf{1}(q_k > 0) + p_k^- \mathbf{1}(q_k < 0)\} \bar{G}(x) = c_k(q_k) \bar{G}(x).$$

Applying Theorem 4.4 to $\sum_k Y_k = q^\top Z$ yields the first-order tail claim. Applying the independent CdMC construction and BRE proof of Appendix B to the K -dimensional shock vector gives the estimator claim, with K replacing N in the BRE bound. $\square$

Proof of Theorem 5.8. By hypothesis the blocks are independent, so the block sums $Y_1, \dots, Y_K$ are independent. By assumption $\mathbb{P}(Y_k > x) \sim d_k \bar{G}(x)$ and the block-level single-big-jump conditions hold. Theorem 4.4 applied to $\sum_k Y_k$ gives

$$\mathbb{P}(S_N > x) = \mathbb{P}\left(\sum_k Y_k > x\right) \sim \sum_k \mathbb{P}(Y_k > x) \sim \left(\sum_k d_k\right) \bar{G}(x).$$

$\square$

Proof of Theorem 6.4. By multivariate regular variation,

$$\frac{\mathbb{P}(X/x \in A)}{\bar{H}(x)} \rightarrow \nu(A)$$

for every Borel set A bounded away from the origin with $\nu(\partial A) = 0$. Take $A = A_\ell = \{z : \ell(z) > 1\}$; this set is bounded away from the origin and admissible by the boundary-continuity hypothesis of Assumption 6.3. Then

$$\mathbb{P}\{\ell(X) > x\} = \mathbb{P}(X/x \in A_\ell) \sim \bar{H}(x) \nu(A_\ell).$$

For the polar representation,

$$\nu(A_\ell) = \int \int \mathbf{1}\{r\ell(\theta) > 1\} \alpha r^{-\alpha-1} dr S(d\theta).$$

If $\ell(\theta) \leq 0$ the inner integral is zero. If $\ell(\theta) > 0$ the inner integral equals $\int_{1/\ell(\theta)}^\infty \alpha r^{-\alpha-1} dr = \ell(\theta)^\alpha$. Integrating over θ gives the stated spectral integral. $\square$

Proof of Proposition 6.5. Condition on Θ . If $\ell(\Theta) \leq 0$, then $\ell(R\Theta) > x$ is impossible for $x > 0$. If $\ell(\Theta) > 0$, the event is $R > x/\ell(\Theta)$. Thus

$$\mathbb{P}\{\ell(X) > x \mid \Theta\} = \bar{F}_R(x/\ell(\Theta) \mid \Theta) \mathbf{1}\{\ell(\Theta) > 0\}.$$

Taking expectations yields the population identity and proves unbiasedness of $Z^{\text{spec}}(x)$. $\square$

Proof of Proposition 6.6. Let $Y = \ell(\Theta)_+$. The envelope assumption gives

$$Z^{\text{spec}}(x) \leq K\overline{H}(x)Y^\alpha,$$

hence

$$\mathbb{E}[Z^{\text{spec}}(x)^2] \leq K^2\overline{H}(x)^2\mathbb{E}Y^{2\alpha}.$$

The MRV linear-risk asymptotic gives $\mu(x) \sim \overline{H}(x)\mathbb{E}Y^\alpha$ under the same normalization. Therefore

$$\limsup_{x \rightarrow \infty} \frac{\text{Var } Z^{\text{spec}}(x)}{\mu(x)^2} \leq K^2 \frac{\mathbb{E}Y^{2\alpha}}{(\mathbb{E}Y^\alpha)^2} < \infty,$$

which is bounded relative error provided $\mathbb{E}Y^{2\alpha} < \infty$. $\square$

D Second-Order and Hidden-Regular-Variation Details

This appendix collects the proofs and auxiliary lemmas for the second-order and hidden-cone expansions. We use the notation of Appendix A.

D.1 A primitive condition for the local second-order equivalence

Assumption 3.5 is the local form used throughout the paper. The following classical condition is sufficient.

Lemma D.1 (Sufficient condition for the local second-order equivalence). *Suppose, for each active i ,*

$$\frac{\overline{F}_i(tx)/\overline{G}(x) - c_it^{-\alpha}}{A(x)} \rightarrow c_id_it^{-\alpha} \frac{t^\rho - 1}{\rho}$$

locally uniformly for t near one, with $\rho \leq 0$, and assume Potter bounds for $\overline{F}_i$ and $\overline{G}$ permitting the local shift $t = 1 - u/x$ for bounded u . Then $\overline{F}_i$ satisfies Assumption 3.5 with the displayed constants c_i, d_i , auxiliary $A(x)$, and the expansion

$$t^{-\alpha} = 1 + \frac{\alpha u}{x} + O(u^2/x^2)$$

for bounded u .

Proof sketch. Substitute $t = 1 - u/x$ in the displayed limit, use the Taylor expansion of $t \mapsto t^{-\alpha}$ at $t = 1$, and apply uniform Potter bounds to swap limit and substitution. The resulting expansion is precisely the local form in Assumption 3.5 with the same c_id_i . $\square$

D.2 A binomial-type KL lemma

The Bernstein-style concentration inequalities for CdMC estimators rely on the following Taylor expansion of the binomial KL divergence.

Lemma D.2 (Binomial KL Taylor expansion). *For $p \in (0, 1)$ and t such that $p + t \in (0, 1)$, the binomial Kullback–Leibler divergence*

$$D_{\text{KL}}(p + t \parallel p) = (p + t) \log \frac{p + t}{p} + (1 - p - t) \log \frac{1 - p - t}{1 - p}$$

satisfies

$$D_{\text{KL}}(p + t \parallel p) = \frac{t^2}{2p(1 - p)} + O\left(\frac{|t|^3}{p^2(1 - p)^2}\right)$$

as $t \rightarrow 0$. For p bounded away from $\{0, 1\}$ the remainder is $O(t^3)$; for $p \rightarrow 0$, the leading constant is $1/(2p(1 - p)) \sim 1/(2p)$ and the cubic remainder is bounded by $C|t|^3/p^2$ for a universal C .

Proof. Set $g(t) = D_{\text{KL}}(p + t \parallel p)$. Direct differentiation gives $g(0) = 0$, $g'(0) = 0$ (by stationarity at $t = 0$), and $g''(0) = 1/(p(1 - p))$. The Taylor expansion to second order yields $g(t) = t^2/(2p(1 - p)) + O(|t|^3 g'''(0))$. The third derivative is $g'''(t) = (p + t)^{-2} - (1 - p - t)^{-2}$, bounded by $2/(p^2) + 2/(1 - p)^2$ on a neighbourhood of zero; the stated remainder bound follows. $\square$

D.3 Proof of Theorem 7.3

Proof of Theorem 7.3. Write $A_S = \{z : \sum_i z_i > 1\}$ and decompose the rescaled event $\{X/x \in A_S\}$ into the axis region $A_S \cap \bigcup_i N_\delta(\{te_i : t > 0\})$ and the hidden-cone region $A_S \cap \mathcal{E}_2$, where $N_\delta(\cdot)$ denotes a δ -tube around each axis. The axis tube width $\delta = \delta(x)$ is chosen so that $\delta(x) \downarrow 0$ sufficiently slowly to retain the localization but fast enough to keep the hidden region bounded away from the axes in the limit.

Axis contribution. On the axis tubes the analysis reduces to the independent second-order calculation of Theorem 4.6, applied to each component. By Assumption 4.3 and Assumption 3.5, this contributes

$$\overline{G}(x) \nu_1(A_S) + \overline{G}(x) A(x) D_{\text{marg}} + \frac{\alpha \overline{G}(x)}{x} D_{\text{mean}} + o(r(x)),$$

where $\nu_1(A_S) = C_{\text{axis}}$, $D_{\text{marg}} = \sum_{i \in \mathcal{A}} c_i d_i$, and $D_{\text{mean}} = \sum_{i \in \mathcal{A}} c_i \mu_{-i}$.

Hidden-cone contribution. On the hidden region $A_S \cap \mathcal{E}_2$, which is bounded away from the axes, Definition 7.1 and Assumption 7.2 give

$$\mathbb{P}(X/x \in A_S \cap \mathcal{E}_2) \sim \overline{H}_2(x) \nu_2(A_S \cap \mathcal{E}_2) = \overline{H}_2(x) D_{\text{hid}}.$$

Overlap. The overlap between the axis tube and the hidden region has mass $o(\overline{G}(x))$ under ν_1 (it lies in $\mathcal{E}_2$, where ν_1 vanishes) and $o(\overline{H}_2(x))$ under ν_2 (it lies near the axes, where ν_2 is asymptotically negligible by the homogeneity and the choice of $\delta(x)$). Triple and higher cones not modelled explicitly are $o(r(x))$ by Assumption 7.2.

Adding the four contributions gives the displayed expansion. $\square$

D.4 Threshold ordering in empirical work

The hidden-cone term is empirically relevant only when

$$\widehat{H}_2(x)\widehat{D}_{\text{hid}} \gtrsim \max\left\{\widehat{G}(x)|\widehat{A}(x)|, \widehat{G}(x)/x\right\}.$$

The data pipeline reports this comparison directly. If the hidden term is below both marginal second-order scales over all forecast thresholds, hidden-cone estimators are reported as diagnostics but excluded from primary VaR/ES forecasts.

E Algorithms and Pseudo-Code

Reference implementation. Every algorithm in this appendix has a one-to-one Python entry point in the `FactorTail` library (<https://github.com/osolari/FactorTail>). Module paths follow the manuscript: `factortail.cdmc.independent`, `factortail.cdmc.dependent`, `factortail.cdmc.latent_shock`, `factortail.cdmc.block`, `factortail.cdmc.spectral`, and `factortail.real_data.rolling_var_es`. Pseudo-code below uses mathematical notation; the implementation respects the numerical-stability conventions of Remark 5.9.

Algorithm 1 Dependent selected-maximum CdMC

Require: threshold x , joint sampler for X , conditional kernels $p_i(t; X_{-i})$

- 1: **for** replicate $m = 1, \dots, n$ **do**
 - 2: Draw $X^{(m)}$ from the fitted joint model.
 - 3: For each i , compute $S_{-i}^{(m)}$, $M_{-i}^{(m)}$, and $T_i^{(m)} = (x - S_{-i}^{(m)}) \vee M_{-i}^{(m)}$.
 - 4: Set $Z_m = \sum_{i=1}^N p_i(T_i^{(m)}; X_{-i}^{(m)})$.
 - 5: **end for**
 - 6: **return** $\widehat{\mu} = n^{-1} \sum_m Z_m$, sample variance, confidence interval, runtime.
-

Complexity and numerical stability (Algorithm 1). Per replicate: $O(N)$ conditional-kernel evaluations plus the cost of one joint draw. Total work for n replicates is $O(n(N c_p + c_{\text{draw}}))$, where c_p is the kernel evaluation cost and c_{draw} is the joint-draw cost. Following Remark 5.9, evaluate kernels on log-scale when the deep tail is involved and apply log-sum-exp when aggregating.

Algorithm 2 Latent-shock CdMC

Require: factor loading matrix B , exposure a , independent shock tails, threshold x

- 1: Compute shock exposure $q = B^\top a$.
 - 2: **for** replicate $m = 1, \dots, n$ **do**
 - 3: Draw all shocks except each candidate shock as needed for summed or stratified CdMC.
 - 4: Compute signed shock thresholds for $q_k Z_k$.
 - 5: Evaluate one-dimensional shock survival functions with sign correction.
 - 6: **end for**
 - 7: **return** tail estimate, attribution by shock, and BRE diagnostics.
-

Complexity and numerical stability (Algorithm 2). Per replicate: $O(K)$ univariate shock-survival evaluations plus $O(K)$ shock draws, where K is the number of latent shocks. Total work is $O(n K c_{F_Z})$ for univariate-survival cost c_{F_Z} . Sign correction for $q_k < 0$ requires evaluating the negative shock tail; both tails should be evaluated on log-scale when the shock distribution has very heavy tails (e.g., α close to 1).

Algorithm 3 Spectral/radial CdMC

Require: angular sampler or empirical angular resample, radial survival model, loss vector a , threshold x

- 1: **for** replicate $m = 1, \dots, n$ **do**
 - 2: Draw or resample angle $\Theta^{(m)}$.
 - 3: Compute $y_m = (a^\top \Theta^{(m)})_+$.
 - 4: **if** $y_m = 0$ **then**
 - 5: Set $Z_m = 0$.
 - 6: **else**
 - 7: Set $Z_m = \mathbb{P}(R > x/y_m \mid \Theta^{(m)})$.
 - 8: **end if**
 - 9: **end for**
 - 10: **return** $\hat{\mu} = n^{-1} \sum_m Z_m$, spectral attribution, and uncertainty bands.
-

Complexity and numerical stability (Algorithm 3). Per replicate: one angular draw or resample plus one radial-survival evaluation at x/y_m . Total work is $O(n(c_\Theta + c_{F_R}))$. When y_m is small, the quantile x/y_m is extreme and the radial survival should be evaluated on log-scale. The estimator $Z_m = 0$ branch when $y_m = 0$ is exact for $x > 0$.

Algorithm 4 Real-data rolling VaR/ES protocol

Require: returns, factors, portfolio definitions, rolling window w , forecast levels τ

- 1: **for** forecast date t **do**
 - 2: Fit rolling factor model on $[t - w, t - 1]$.
 - 3: Construct signed factor and residual loss contributions.
 - 4: Estimate marginal tails and dependence diagnostics.
 - 5: Select candidate estimators using Figure 6.
 - 6: For each estimator and τ , solve for $\widehat{\text{VaR}}_t(\tau)$ and compute $\widehat{\text{ES}}_t(\tau)$.
 - 7: Record forecasts, runtime, model diagnostics, and provenance.
 - 8: **end for**
 - 9: Run VaR/ES backtests and write validated tables/figures.
-

Complexity and numerical stability (Algorithm 4). Per forecast date: one rolling fit ($O(w d^2)$ for a d -factor OLS), tail fitting ($O(w \log w)$ for order statistics), dependence diagnostics ($O(w N^2)$ for pairwise tail measures), and one CdMC run per selected estimator (work as above). For a T -date backtest the total work scales as T times the per-date work. VaR and ES are computed by root-finding on $\hat{\mu}(\cdot; \theta_t)$; the root-finding tolerance should be tighter than the estimator's Monte Carlo half-width.

F Data Specifications

F.1 Public factor and portfolio files

The public pipeline downloads or records local copies of these files:

- daily Fama–French three factors;
- daily Fama–French five factors;
- daily momentum factor;
- daily industry portfolios;
- daily size/book-to-market portfolios;
- daily profitability and investment portfolios when available;
- risk-free rate series bundled with the factor files.

For each file the pipeline stores source URL, data vintage, access date, row count, first date, last date, checksum, and parsing log.

Vintage and citation requirements. The Kenneth French Data Library is updated periodically and is the primary public source [French, 2026]. Each replication run records the `Last_modified` field of every source file, the download timestamp, and the SHA-256 file hash. Citations in the final paper must reference both the data-library entry and the specific vintage used. When the data are reissued upstream (e.g., factor revisions), the manuscript run is not retroactively updated; new vintages create new runs.

F.2 Licensed CRSP extension

The licensed extension requires CRSP daily stock returns through WRDS. The planned filters are ordinary common shares (`shrcd` $\in$ $\{10, 11\}$), NYSE/AMEX/NASDAQ exchanges (`exchcd` $\in$ $\{1, 2, 3\}$), nonmissing returns with delisting-return integration following standard CRSP guidance, and market-equity screens. Security-level portfolios are formed after all filters are recorded. *No CRSP data are included in this archive.*

License and citation. CRSP data are licensed by the Center for Research in Security Prices at the University of Chicago Booth School of Business and are subject to the WRDS end-user license agreement [Center for Research in Security Prices, 2026]. Replication of the CRSP-based results requires institutional WRDS access. Each CRSP-derived output records the WRDS access date and the CRSP file version (`dsf_yyyyymm`).

F.3 Output datasets

All generated outputs are written under `results/`. Required columns are listed in `results/SCHEMA.md`. The schema and writers are implemented in the `factortail.io` module of the `FactorTail` library (<https://github.com/osolari/FactorTail>). Each output contains:

- `run_id`, `config_hash`, `git_hash`;

- data source and vintage;
- random seed and spawned seed where applicable;
- estimator name and model class;
- threshold or forecast level;
- estimate, standard error, confidence interval, and runtime;
- validation status.

G Experiment Manifest and Replacement Rules

The replacement rules in this appendix are enforced by the `factortail.manifest` module of the `FactorTail` library (<https://github.com/osolari/FactorTail>); the CLI command `factortail validate-run` checks each rule before allowing a generated artifact to replace a placeholder in the manuscript build.

G.1 Figure replacement rule

Each placeholder figure contains a source CSV name. A placeholder may be replaced only when:

1. the CSV exists and passes schema validation;
2. the generation script records run ID, config hash, git hash, and seed;
3. the generated PDF has the same basename as the placeholder;
4. the caption and label in the manuscript remain unchanged unless the underlying diagnostic changes.

G.2 Table replacement rule

Each table with `xx` cells is a planned table. Replacement requires a generated `.tex` table with the same columns and a validation log. The validation log must state whether each value is empirical, analytic, simulated, or manually entered.

G.3 Single manifest aligned with the experiment-status table

The priority manifest below corresponds, row-for-row, to the seven experiments in Table 8. Each row names the experiment, the required generated outputs, the labels of the figures and tables that the outputs replace, and the current run status. The replacement rules above govern how each output enters the manuscript.

Priority	Experiment	Required outputs and target labels	Required CSVs	Status
P1	Independent replication	F1 (Figure 3), F8 (Figure 4), Table 6	F1, sim_indep	F8, planned

Priority	Experiment	Required outputs and target labels	Required CSVs	Status
P2	Common-shock simulation	F11 (Figure 5), Table 7 rows for Family II	F11, sim_common_shock	planned
P3	Copula-kernel test	Table 7 rows for Family IV	sim_copula	planned
P4	MRV spectral test	F12 (Figure 8), Table 7 rows for Family V	F12, sim_spectral	planned
P5	Hidden-cone test	F13 (Figure 9), Table 7 rows for Family VI	F13, sim_hrv	planned
P6	Public Fama–French data	F15–F18, Tables 10 to 14	ff_panels, ff_tails, ff_dep, ff_backtest, ff_crisis	planned
P7	CRSP licensed extension	F15–F16 with CRSP universe; security-level panels in Table 10	crsp_panels, crsp_tails, crsp_dep	optional/planned

The mapping above is authoritative: any new experiment must be added both to Table 8 and to this manifest before being run, to preserve the one-to-one correspondence.